PATENT APPLICATION

Topologically Protected Recording Device with Sectored Architecture and Multi-Stage Signature Chain

TECHNICAL FIELD

The present invention relates to a recording device for generating tamper-resistant audiovisual data, in particular a recording device having a sectored architecture in which a proprietary first sector and an open second sector are unidirectionally connected by a physical data diode, as well as associated methods and platform systems for authenticity verification.

BACKGROUND OF THE INVENTION

Problem Statement

Due to the rapid development of generative artificial intelligence, humans are increasingly unable to distinguish genuine recordings from AI-generated images, videos, or audio content. Deepfake technologies enable the creation of photorealistic content that is practically indistinguishable from authentic recordings. This poses a fundamental threat to trust in visual and audiovisual media, with severe implications for journalism, war reporting, surveillance, jurisprudence, and societal communication.

Prior Art

Known approaches include:

In-camera signing (Sony Camera Authenticity Solution, C2PA): Sony implements an in-camera digital signature based on the C2PA standard, in which a signature is generated at the time of capture and embedded in the image. The keys are stored in a hardware chipset. Additionally, 3D depth information is used for detecting screen re-recordings. However, the architecture is monolithic: a single security domain encompasses all cryptographic functions, without topological separation between the manufacturer domain and the user domain.

Blockchain-embedded camera systems (Columbia University, WO2019236470A1): This system combines a digital camera with a security module (HSM) and blockchain-capable hardware. Hashes of the audiovisual material are published via smart contracts on a public blockchain. GNSS data and spectral analysis serve for location and time verification. This system is also monolithically structured and does not feature sectorization. Tamper protection is achieved through a physical tamper detection box and GPIO-based detection of unauthorized connections.

Multi-stage signature chains via separate devices (Skydio, US12231575B2): Skydio describes an architecture for Trusted Contextual Content (TCC), in which multiple physically separate devices - for example a drone and its controller - each generate their own signatures and chain them together. Each device signs the entire accumulated data

package including prior signatures. However, the signatures are distributed across separate physical devices, not across topologically isolated sectors within a single recording device. A unidirectional hardware data diode between the signature instances is not provided.

PUF-based key generation in image sensors: It is known to integrate Physical Unclonable Functions (PUFs) in image sensors in order to generate unique cryptographic keys from manufacturing-induced variations (US11652649B2, US11316705B1). However, these approaches are limited to single-device architectures and do not combine PUF with a sectored dual-signature architecture.

Data diodes: Hardware-based unidirectional data transmission systems are known in the field of IT/OT security. They are used in critical infrastructures for network segmentation. However, their use within a recording device for topological separation of cryptographic domains is not known.

Further individual aspects of the prior art: LegitIPix (US10439821B2) describes a direct signature embedding in pixel data of a CMOS sensor using a shared-key approach. NXP (US20250202718A1) discloses an eIDAS-based certificate provisioning via a provisioning device. Amazon (US11770260B1) teaches a frame-by-frame hash chain for video verification. Basler (US20210385408A1) discloses an authentication module for sensor data with steganographic modification. EP3262782B1 describes an anti-tamper system with decoupled architecture and zeroization.

What the known solutions have in common is that they feature a monolithic security architecture in which a single trust domain encompasses both the sensor system and the entire cryptographic processing. This means that the manufacturer or a single attacker with access to this trust domain can potentially compromise the entire system. There is therefore a need for an architecture that substantially increases tamper resistance through topological separation and physically, i.e., hardware-enforced, unidirectionality.

SUMMARY OF THE INVENTION

The present invention solves the stated problem by a recording device having a sectored architecture, comprising:

a) a first sector, which comprises at least one sensor and a first signature engine, wherein the first signature engine is configured to determine cryptographic hash values from a signal captured by the sensor, and to digitally sign these with a private key of a key pair assigned to the first sector, in order to generate a first signature;

b) a second sector, which comprises at least a second signature engine, wherein the second signature engine is configured to receive the signal and the first signature and to generate a second signature using a private key of a key pair assigned to the second sector;

c) a data diode arranged between the first sector and the second sector, which effects a physically enforced unidirectional data transmission from the first sector to the second sector and prevents a reverse flow of information from the second sector to the first sector;

wherein the first sector and the second sector are assigned to different trust domains.

DETAILED DESCRIPTION

The present invention is described below by way of example with reference to a video camera. This example is non-limiting; the concept is transferable to all recording devices, including audio recorders, surveillance cameras, body cameras, and smartphones.

1. Sectored Architecture

1.1 First Sector (Proprietary Sector)

The first sector is attributable to the manufacturer of the recording device and is preferably designed to be as manufacturer-specific as possible. It is preferably arranged entirely or partially within the housing of the recording device and, for example, entirely or partially within the second sector. Within the first sector there are at least:

- an image sensor for capturing visual data;
- optionally further sensors, in particular microphones for capturing acoustic data;
- a first signature engine, which generates cryptographic hash values from the audiovisual signal and signs these with a private key of a key pair assigned to the first sector.

The protection of the first sector is achieved primarily through secrecy and manufacturer-specific design. Preferably, the first signature engine is implemented as a System-on-Chip (SoC) directly on the chip of the image sensor, in order to minimize the path between the data source and the signing point. The signing is performed close to the source. Preferably, not only the raw data delivered by the image sensor and any microphones, but also metadata generated in the first sector - in particular autofocus information, depth sensor data, liveness parameters, and/or other sensor data of the first sector - are included in the hash value computation before leaving the first sector and are jointly signed there. Preferably, the cryptographic hash values are determined on a per-frame basis.

1.2 Second Sector (Open Sector)

The second sector is attributable to the user of the recording device. In contrast to the first sector, protection here is achieved not through secrecy, but through maximum transparency. Preferably, the second sector is designed as an open-source system whose hardware and software are publicly viewable and verifiable. Preferably, the

second sector is also arranged entirely or partially within the housing of the recording device. Within the second sector there is at least a second signature engine. This receives, via the data diode, the audiovisual signal together with the first signature of the first sector and generates a second signature. Preferably, in the second sector, further metadata are added to the audiovisual signal, which are collected by means of additional sensor systems, in particular:

- a GNSS module for position determination;
- a long-wave time signal receiver (e.g., DCF77 at 77.5 kHz) for an independent time reference;
- inertial sensors including gyroscopic sensors and acceleration sensors;
- additional environmental sensors.

1.3 Data Diode

A data diode is arranged between the first and the second sector. The term data diode is to be interpreted broadly and encompasses any device that effects a physically enforced unidirectional data transmission. The data diode may in particular be realized by:

- optoelectronic couplers (optocouplers), in which a light-emitting component in the first sector drives a light-sensitive component in the second sector, without a practically usable reverse channel physically existing (the parasitic capacitance of the optocoupler must be taken into account by a person skilled in the art, as it opens up a hypothetical back-action possibility);
- fiber-optic connections having exclusively one light-transmitting and one light-receiving unit;
- circuits with strongly asymmetric transmission characteristics, in which signal transmission in the reverse direction is physically prevented or attenuated to such an extent that a reverse flow of information is not possible.

It is decisive that at the interface between the sectors, data can flow exclusively in one direction - from the first to the second sector. Thereby, an attacker who has compromised the second sector is physically prevented from acting upon the first sector across the data diode.

2. Signature Chain

2.1 First Signature (Proprietary Sector)

In the first sector, cryptographic hash values are determined from the audiovisual signal. Preferably, this is performed on a per-frame basis, i.e., for each captured frame a separate hash value is generated. The hash values are signed with the private key of the first sector, thereby producing the first signature. It is possible to compress the recording

data prior to the hash value computation. This may also be performed in parallel: In the first sector, a RAW stream may be generated and signed on a per-frame basis, and from the RAW stream, reversibly or irreversibly compressed streams may be generated, which are likewise independently signed. The data diode then forwards both the RAW stream and the compressed streams together with their respective signatures to the second sector.

2.2 Second Signature (Open Sector)

In the second sector, the received data package - consisting of the data package already fully signed by the first sector (audiovisual data and, where applicable, metadata of the first sector) and the first signature - is subjected to a second signing together with the metadata collected in the second sector. The second signature engine generates cryptographic hash values over the total data package and signs these with the private key of the second sector.

The dual signature chain ensures that a forgery requires the compromise of both independent sectors. Since these are assigned to different trust domains (manufacturer vs. user) and are physically separated by the data diode, tamper resistance is substantially increased compared to monolithic systems.

3. Key Generation and Key Management

3.1 PUF-Based Key Generation

In a preferred embodiment, the generation of the private keys is performed with the involvement of a Physical Unclonable Function (PUF). A PUF utilizes manufacturing-induced, unavoidable variations in the semiconductor structure in order to generate a unique, hardware-bound entropy contribution. The private key of each sector preferably comprises at least two components:

- a PUF-derived component, which is obtained from the physical structure of the respective sector;
- a manufacturer-side or user-side component, respectively, which is introduced during initialization.

Through this partitioning, neither the manufacturer alone nor the user alone knows the complete private key of their sector. Additional "key sources" may be added, wherein those are preferred in which collusion with the manufacturer or user is implausible. The PUF hardware itself also does not "know" the complete key, since it merely provides an entropy contribution. Preferably, after the PUF hardware has delivered its entropy contribution, it is irreversibly altered, which in systems based on MOS structures can be achieved by applying an otherwise impermissible gate-source voltage V_GS. In this process, preferably a sufficiently dimensioned voltage pulse is applied to destroy all gates of the MOS structure such that even electron-microscopic examinations no longer

provide any indication of a prior memory state. It is thus achieved that after initialization of the recording device, the private keys exist exclusively in volatile memories, specifically in read-protected memories of the camera, and therein in at least two disjoint memories (namely at least one in the first and at least one in the second sector). The described method, in which a key is determined or obtained from an external contribution and a contribution of a PUF, and is then stored in a volatile memory, whereupon the PUF is irreversibly altered, is not limited to recording devices, but may be used in all applications in which the loss of a hardware-bound key causes lesser costs than a theft thereof.

3.2 Key Storage

The private keys are preferably held in volatile memory (e.g., SRAM) that is permanently powered by a battery. A loss of supply voltage leads immediately to the loss of the keys. In conjunction with the anti-tamper measures described below, comprehensive protection against physical key extraction is thereby achieved.

4. Tamper Protection (Anti-Tamper)

4.1 Gas Monitoring

In a preferred embodiment, at least the first sector of the recording device is filled with a gas, preferably an inert gas, whose pressure deviates substantially from the ambient pressure. During the entire service life of the recording device, the pressure and temperature of the gas are measured quasi-continuously. A memory erasure (zeroization) is triggered as soon as one of the following conditions is detected:

- an excessive rate of pressure change, indicating an opening of the housing;
- an absolutely too high or too low pressure;
- an excessive deviation of the sensor data from the ideal gas law, which may indicate a change in the effective sector volume (opening) or a manipulation of the sensor system itself;
- an impermissibly low temperature, in particular for protection against cold boot attacks, in which an attacker attempts to preserve the volatile memory at low temperature and to read it out.

The pressure measurement is preferably temperature-compensated in order to avoid false alarms (and consequently bricking) due to temperature changes during normal operation. The described gas monitoring also provides protection against non-thermal data remanence attacks such as, e.g., "volt boot" attacks, since it substantially impedes successful physical access to the key memory. Additionally or alternatively, monitoring of the chemical composition of the gas fills is possible, for example with electrochemical or optical sensors, wherein the precise composition of a gas fill of the manufacturer is preferably a manufacturer secret. The described gas monitoring with

pressure-dependent memory erasure is not limited to recording devices, but may be employed in any electronic device that is intended to protect a cryptographic key held in volatile memory against physical access.

4.2 Thermal and Chemical Destruction

Preferably, upon detected tampering, not only is the memory content erased, but in addition a thermal destruction of the memory cells is initiated, for example by short-circuiting a battery arranged in the respective sector for maintaining the memory state. For example, the waste heat of a fuse associated with a battery may be used directly or indirectly (e.g., by igniting a reactive composite such as Nanofoil™) for thermal destruction. The fuse itself may also comprise a reactive composite ("Pyrofuze"). Thereby, the memory cells and their contents are irreversibly destroyed. Additionally or alternatively, destruction of the memory by chemical reaction with atmospheric constituents may be provided, which commences upon loss of housing integrity.

4.3 Further Tamper Protection Measures

A combination with further known forms of tamper protection is recommended, including tempered glasses, sensor wires, conductive mesh structures, and similar means.

5. Metadata and Plausibility Checking

5.1 Long-Wave Time Signal and GNSS Cross-Validation

Particularly advantageous is the combination of a long-wave time signal receiver and a GNSS module within the second sector. This enables an immediate plausibility check:

- Comparison of timestamps: The long-wave time signal (e.g., DCF77) provides a time reference independent of satellite navigation. A comparison with GNSS time permits the detection of GNSS spoofing.
- Comparison of the location defined by GNSS data with the long-wave time signal, taking into account the propagation time from the transmitter to the camera: If the location of the long-wave transmitter (e.g., Mainflingen for DCF77) is known, the approximate distance to the transmitter can be inferred from the reception time and the signal propagation time, which must be consistent with the GNSS location.

5.2 Atmospheric Environmental Signals as Fingerprints

The antenna of the long-wave time signal receiver may additionally be used to measure unpredictable natural atmospheric signals that occur independently of electrical grids, in particular:

- "Radio Hiss", VLF hiss (Very Low Frequency noise);
- Sferics (electromagnetic impulses from lightning discharges);

- Tweeks (dispersive long-range propagation of sferics in the Earth-ionosphere waveguide structure);
- Fluctuations in the signal strength of the time signal transmitter.

In the case of a live broadcast to the Internet, it would be immediately apparent if, e.g., the VLF hiss of one camera deviates from that of similar cameras in the vicinity. For verification, in particular of sferics, observation infrastructures may be consulted, such as in particular Earth observation satellites, networks of geophysical sensors, or lightning location networks.

5.3 Inertial Sensors for Consistency Checking

Inertial sensors (acceleration sensors, gyroscopic sensors) in the second sector capture the motion of the recording device. By comparing the camera motion with the image content, plausibility can be verified: if the camera rotates but the image remains stable (or vice versa), this indicates tampering.

6. Anti-Screen-Recording (Protection Against Re-Recording from a Display)

A further aspect of the invention relates to protection against re-recording of falsified images or videos from a display, wherein even lens attachments with built-in screens would be conceivable. Against this, the following measures are proposed in particular:

- depth sensors, wherein preferably known or existing autofocus systems are incorporated;
- multiple systems, in particular also those with active object illumination;
- systems that interact with the object to be recorded via ultrasound;
- liveness sensors, in particular for recording humans.

The image sensor itself may serve as a liveness sensor if it is designed as a 4-channel sensor (RGB-IR sensor). Otherwise, an additional infrared-sensitive image sensor may be provided. Their data are not only used for color correction, but are treated as part of the recording. The signals of the above-mentioned sensors are fundamentally accorded the same significance as those of the image sensor and the microphones, and are likewise signed close to the source. In this context, imaging sensors are preferably assigned to the first sector, while illumination means are assigned to the second sector.

Illumination means in the second sector may not serve exclusively for screen-recording protection, but may transmit data to an image sensor of the first sector via object illumination, which data are then first signed in the first sector as part of the recording. A modulatable light source of the second sector may, for example, be both part of a known active autofocus system and, e.g., by amplitude or directional modulation, carry data useful for subsequent authentication of the recording, which data become "visible" in the recording (optionally also in the infrared range, in particular NIR). These data may

comprise, e.g., unencrypted or encrypted (with a private key of the second sector) geolocation data including GNSS time, otherwise obtained time, identity of the camera, and an environmental fingerprint (in particular a radio fingerprint, and therein preferably a VLF fingerprint). Since the first sector does not interpret the data contained in the modulated light, but merely captures and signs them as part of the audiovisual signal, the informational isolation of the first sector is preserved; a falsification of the modulated data by a compromised second sector would be detectable upon external verification.

The latter concept of information-carrying object illumination may be extended: A device for object illumination may be presented independently of the present invention according to claim 1, which device serves as an accessory to existing cameras with C2PA support to improve trust in their recordings by illuminating objects to be recorded in a correspondingly "data-carrying" manner; the recordings may then contain, e.g., a C2PA-signed environmental fingerprint, preferably a VLF and/or shortwave fingerprint.

7. Blockchain Registration and Platform Integration

7.1 Publication of Public Keys

The manufacturer publishes the public key of the first sector in a practically unforgeable manner, in particular in a blockchain, wherein metadata of the camera, in particular a digital fingerprint of its first sector, may be associated with the public key.

This fingerprint may comprise a mapping of the pixel defects of the image sensor. Additionally or instead of a pixel defect mapping, the fingerprint may comprise a signature derived from the Photo-Response Non-Uniformity (PRNU) of the image sensor (PRNU describes the manufacturing-induced, unique spatial pattern of slight sensitivity differences of individual pixels for each image sensor, which can be statistically extracted from a plurality of recordings, and which enables forensic identification of the image sensor even when the cryptographic key has been compromised.) The forensic value of such fingerprints uniquely associated with an image sensor and published in a practically unforgeable manner would be immense.

As regards the user, the user is enabled to likewise publish the public key of the second sector (comprising the open-source signature engine) in the blockchain, and to cryptographically link it unambiguously with the public key of the manufacturer.

7.2 Authenticity Verification

For authenticity verification, a cryptographic hash value is newly computed from the present audiovisual material including metadata. Then the first signature is verified by means of the public key of the first sector deposited in the blockchain, and the second signature is verified by means of the deposited public key of the second sector. The successful verification of both signatures confirms both the integrity and the authenticity of the recording.

7.3 Platform and Browser Integration

In a further embodiment of the invention, video platforms, social media platforms, web browsers, or browser applications are proposed which check, for each uploaded alleged recording (audio, video, image), whether it can be attributed to a recording by a recording device that digitally signs close to the source through an intact certificate chain. The platform signals this to the user in an understandable manner or enables the user to preferably receive only recordings verified as authentic.

8. Further Notes on Protection Against Attacks

8.1. Cross-Sector Key Erasure

Preferably, both the first and the second sector are quasi-continuously tamper-protected to the effect that, in the case of all assumed tampering attempts, an immediate erasure of the respective private keys is achieved. Further preferably, triggering of a tamper protection device in one sector even effects an erasure of all private keys in all camera sectors, however without digital information crossing sector boundaries. This may be achieved, for example, thermally or mechanically, in that a fuse of a battery which powers the volatile key memory in one sector necessarily co-triggers one or more corresponding fuses assigned to other sectors. For this purpose, for example, a simply connected Al/Pd wire ("Pyrofuze") may be used, in which a first section in the first sector and a second section in the second and optionally further sectors serves as an active fuse wire; in this way, the command for key erasure may cross a sector boundary as a diffusion-controlled chemical reaction without an electrical or electromagnetic signal being transmitted. The described cross-sector key destruction, in which an exothermic reaction front propagates across a sector boundary by means of a reactive element and thereby effects the erasure or destruction of key memories in multiple security domains without an electrical or electromagnetic signal crossing the sector boundary, is not limited to recording devices, but may be employed in any electronic device that has two or more physically separated security domains, each with their own key memories.

8.2. Protection of the Data Diode

According to the invention, the only information-carrying connection between the sectors consists of one or more data diodes. Thus, the inputs and outputs of the data diodes are located in different sectors. Preferably, at least one tamper protection device of one of the sectors is designed to also detect such tampering attempts that concern the transmission path of the data diode, which may be located outside both sectors.

8.3. Signal Paths as Means of Tamper Detection

In recording device systems according to the invention, the image sensor is arranged within the tamper-protected first sector, which itself may be located within the second sector, and can be optically stimulated externally, while physical interference is

impeded. Also, the signal path or paths between the image sensor and the signature engine(s) are tamper-protected within the first sector. These signal paths may be monitored and thereby themselves form part of a tamper detection device.

8.4. Inductive and Capacitive Attacks

For protection against electromagnetic attacks, it is proposed to electromagnetically shield the sectors, and particularly the first sector, in a manner known to a person skilled in the art, with appropriate filters for technically unavoidable feedthroughs. Furthermore, by coating the rear lens element or a protective glass of the camera on one or both sides with an electrically conductive coating (an ITO layer may additionally serve for reflection reduction), the first sector may form a nearly hermetic Faraday cage. The term glass or protective glass is to be interpreted broadly and shall herein include optical glass ceramics. Preferably, such electrically conductive layers are galvanically connected to an electrically conductive housing of the first sector. Insofar as an intended adjustment of a lens or an electromotively executed movement causes a volume change of a sector, a correspondingly present gas (pressure) monitoring system anticipates a corresponding pressure change, so that the gas monitoring system does not trigger. On the contrary, it shall trigger if an inevitably expected pressure change fails to occur.

Insofar as gas monitoring is used, the lens or the protective glass may be connected to the sector housing via a permanent seal, and must be dimensioned to permanently withstand the pressure differential between the sector pressure and the atmospheric pressure. A material-bonded connection between "glass" and camera housing is preferred, which becomes possible by means of active solders or by partial metallization of the glass. In this context, the different thermal expansion coefficients between the optical materials and the housing material must be taken into account by a person skilled in the art, for example by encasing the "glass" in Invar or Kovar and soldering it thereto, and, taking into account possible mechanical stresses, the Invar or Kovar element is in turn connected to the camera housing in a material-bonded manner (by soldering or welding). Furthermore, to avoid common-mode interference within the sectors, including intentional common-mode interference, a consistently differential signal transmission may be used, at least outside of integrated circuits. Finally, attack-vulnerable electronics may be potted, wherein the potting compound may be selected to be insoluble in organic solvents including halogenated solvents, and may contain fillers to influence its thermal conductivity (e.g., AlN) or its electromagnetic properties (e.g., ferrite powder). Further preferably, the potting compound is selected such that all solvents, including supercritical fluids, that are capable of attacking, substantially swelling, or even dissolving it, are also capable of attacking and destroying the potted electronics. In addition to such passive protection, active protection is also possible here: as soon as an electromagnetic attack is detected, for example by an overvoltage event, key erasure is triggered without delay.

8.5. Tamper Protection by Absent Response

Furthermore, at least in the second sector, a key erasure may be triggered as soon as an expected signal from the data diode fails to arrive, since this may be an indication of an attack directed against the first sector. The expected signals from the data diode need not follow a regular interval, but preferably follow one another at pseudo-random intervals.

8.6. Permanent Cryptographic Self-Observation

In a further preferred embodiment, the recording device is configured to generate, during its entire operating life - including outside of regular recording operation - at preferably pseudo-random time intervals, whose expected value preferably lies in the range of seconds to a few minutes, compressed background images of low resolution by means of the image sensor of the first sector. Each background image traverses the regular signature path: it is signed close to the source in the first sector, unidirectionally communicated to the second sector via the data diode, and there subjected to a second signature together with the current metadata - in particular GNSS position, long-wave time signal, radio environmental fingerprint, and inertial sensor data. Additionally, each chain element thus generated is cryptographically chained to the preceding one by incorporating the cryptographic hash value of the preceding chain element into the subsequent one, so that a seamless, forgery-proof image chain over the entire service life of the device is created. The image chain is stored locally in the second sector in a symmetrically encrypted manner, wherein the symmetric key is known outside of the recording device exclusively to the user, and within the recording device is preferably held only in a volatile, read-protected memory. If the authenticity of a recording made with the recording device is disputed, the user may decrypt the entire image chain or - by means of a Merkle tree formed over the chain elements - a verifiable temporal segment thereof and submit it for examination; the integrity of the chain is verifiable through seamless recomputation of the hash values, and any subsequent manipulation, insertion, or deletion of an image destroys the consistency of the chain. By keeping the image resolution sufficiently low and applying sufficiently strong data compression, such a camera can document its integrity on commercially available mass storage media over many years. Object illumination, e.g., NIR object illumination, according to Section 6 may be incorporated into the protection concept.

9. PQC (Post-Quantum Cryptography)

Proceeding from the strategic conjecture that NP is not a subset of BQP, various new encryption and signature algorithms are being developed in anticipation of powerful quantum computers, so-called PQC algorithms. Preferably, at least one signature engine of the first or second sector applies a PQC algorithm. Further preferably, both the first and the second sector use signature engines with PQC algorithms, and preferably different ones, and further preferably ones that are based on different mathematical

problems which cannot, or cannot efficiently, be reduced to one another, in particular selected from: lattice-based problems, code-based problems, hash-based schemes, and/or multivariate polynomial equation systems.

10. Blockchain

In addition to the camera itself, a compromise of the blockchain is also conceivable, in which the public key of the manufacturer is deposited in unambiguous association with that of the user. To this end, it is proposed to deposit these linked key pairs in a plurality of established blockchains, so that an attacker would have to overcome the independent consensus mechanisms of a plurality of blockchains simultaneously.

11. Powering On and Off the First Sector

Insofar as powering on and off the first sector is required, this is preferably performed from the second sector, which preferably surrounds the first, by switching the supply voltage on and off. It shall be ensured by hardware that no changes to the firmware in the first sector can thereby be produced. Preferably, the firmware of the first sector distinguishes between two operating modes based on the duration of the applied supply voltage:

- In the case of a short voltage pulse whose duration is below a predetermined threshold duration, the first sector generates, after booting, a single signed image (challenge-response frame) and outputs it via the data diode to the second sector, whereupon the second sector switches the supply voltage off again.
- In the case of a supply voltage that is applied beyond the threshold duration, the first sector commences regular audiovisual recording operation.

The second sector monitors the response time of the first sector to the voltage pulse. If the expected response fails to arrive within a predetermined time window, or if the response time deviates impermissibly from a time profile characteristic of the first sector, this is treated as a tampering indicator and may trigger a key erasure according to Section 8.5. The challenge-response frames generated at pseudo-random intervals are preferably incorporated into the image chain according to Section 8.6. This serves to counter side-channel attacks.

12. Digital End Devices

The integration of recording devices according to the invention into digital end devices such as smartphones enables anyone to make recordings of the highest evidentiary value.

13. Definitions

A "sector" within the meaning of this application designates a physically delimited functional area within a single device, which comprises its own computing units, its own memory, and its own cryptographic infrastructure, and which is informationally isolated

from other sectors of the same device - that is to say, between the sectors there exists no intended or unintended information path other than the provided data diode, wherein side channels such as electromagnetic radiation, power consumption patterns, thermal coupling, or shared clock sources are suppressed by suitable design measures. A "trust domain" is the totality of those hardware and software components whose integrity is under the control of exactly one responsible entity - such as the manufacturer or the user - and which are associated with a common trust anchor; different trust domains are characterized by the fact that none of them needs to presuppose the integrity of the respective others. A trust anchor is the component within a trust domain whose authenticity and integrity is not derived from a further entity within the same domain, but is assumed as given - typically a cryptographic key or a hardware module harboring such key, from which the chain of trust for all remaining components of the domain originates. A data diode is a physical connection between two sectors that restricts the information flow by hardware to exactly one direction and physically prevents a reverse flow unlike a merely software-based access control. A signature engine designates a unit integrated into the respective sector that forms cryptographic hash values from received data and digitally signs these by means of a private key exclusively assigned to the sector. Close to the source means that the signing is performed spatially and in terms of circuitry so close to the sensor that between the data capture and the signing, no unsecured, externally accessible signal path exists.

14. Network-Capable Cameras and Live Transmission

Recording devices according to the invention may, as already mentioned, be network-capable, for example internet-capable, and may in particular be configured for live transmission of audiovisual content. In a preferred embodiment, the recording device is configured to cooperate with an external provenance or time anchoring protocol which provides for anchoring of cryptographic hash values with one or more device-external services. Such external services comprise in particular Time Stamping Authorities (TSAs) according to RFC 3161, publicly auditable Transparency Logs as provided for by the IETF SCITT architecture, as well as blockchains. Examples of provenance protocols that build upon one or more of these anchoring mechanisms are the C2PA standard in its respective version, the Content Provenance Profile (CPP), as well as the protocols of blockchain-based timestamping services. The invention is not limited to a particular protocol.

The sectored architecture of the invention and such provenance protocols complement each other advantageously: Known provenance protocols establish an externally verifiable time binding of the recordings, but make no statement about the physical integrity of the recording device itself: a device compromise prior to the recording typically lies outside their threat model. The sectoring according to the invention with a physical data diode ensures precisely this device integrity. The combination of both methods thus offers a hitherto unattained level of security, since both the time of the

recording (through the external service) and the authenticity of the recording device (through the dual signature chain) become independently verifiable. This creates trust that the time-stamped and signed apparent recording is in fact a recording and neither a playback nor material generated in quasi-real time.

A provenance protocol core is preferably executed in the second sector (open sector), since this sector may have the requisite network connectivity, and the provenance protocol there processes exclusively data that has already been signed in the first sector and received via the data diode (in addition to the metadata generated in the second sector). An attacker who compromises the provenance process in the second sector cannot forge the first signature of the proprietary sector. Alternatively, the protocol core may be executed outside the second sector, and preferably in a third protected sector, but under no circumstances in the first sector, since the integrity of the first sector must be kept free from network connectivity and the associated attack vectors.

In particular, embodiments of the invention that record radio signals (sferics, tweeks, (VLF) hiss) or information extracted from radio signals according to Section 5.2 as metadata gain considerably in credibility through an additional external time anchoring: The atmospheric radio signal, for example, exists unpredictably only at a specific location at a specific time and cannot be predicted; the at least one external service provides an independently verifiable proof of time. Together, a location-time binding arises that no single actor is able to forge, since the external service cannot know the local radio fingerprint, and a local attacker cannot manipulate the external proof of time.

A single TSA, however, constitutes a single point of failure and a worthwhile attack target whose compromise may go unnoticed. And even a Transparency Log is only as trustworthy as its auditors: To avoid concentrating trust in a single external service, the following improvement is proposed: The recording device preferably connects to a plurality of mutually independent external services, which preferably belong to different categories, for example a TSA and a blockchain, or a Transparency Log and a TSA. The selection of the specific services is preferably made by a transparent random or pseudorandom method, so that the selection is verifiable after the fact but not predictable for an attacker. For example, the pseudorandom selection may be derived from the VLF signal itself, for instance by applying a cryptographic hash function to the most recent VLF environmental fingerprint in conjunction with a publicly known service list.

The external time anchoring provides an upper time bound ("the cryptographic hash value existed no later than at this point in time"). The lower time bound is established by the device-local, offline-capable time references (in particular long-wave time signals, GNSS time, VLF environmental fingerprint). The recording device generates and signs provenance data locally, even without an existing network connection, and buffers this data outside the first sector. When a network connection exists or is established, the buffered data is transmitted to the external services. The recording device is thus

autonomous in its cryptographic core operation; the external anchoring constitutes an additional hardening, not a prerequisite.

Further preferably, the recording device independently captures connection metadata and includes this in the hash value computation. This connection metadata is added to the data packet in the second sector prior to the second signing and is co-hashed. This metadata may comprise in particular the network address, TLS certificate chains of contacted external services, DNS resolution data, and/or results of a traceroute performed by the recording device itself. It provides forensic value, since it can subsequently be correlated with records of the network operators, insofar as such records are available. The cooperation of network operators is, however, not a prerequisite of the method according to the invention, but rather an additional, optional validation possibility.

If a camera according to the invention has self-collected GNSS data, and its clock is cross-validated by a time signal transmitter, and in addition a VLF environmental fingerprint is collected and co-signed, a highly reliable proof of provenance can be conducted entirely without time anchoring by external services, provided that the VLF environmental fingerprint can be reconstructed. Since a sufficiently dense and forensically robust VLF reference infrastructure is not yet available on a comprehensive basis at this time, the external time anchoring according to the foregoing embodiments represents, for the time being, the practically preferred method of external time binding, while the VLF environmental fingerprint serves as a complementary, offline-capable, and third-party-independent proof of provenance.

15. Recovery from Bricking (Bricking Recovery)

Insofar as a recording device according to the invention has a tamper protection according to Section 4, which upon detected or suspected tampering triggers an immediate erasure of the private keys held in volatile memory, which is strongly recommended with regard to system security, an erroneous triggering thereof leads to unintentional bricking, i.e., an irreversible loss of the cryptographic identity of the recording device. Such an erroneous triggering of the tamper protection would inevitably also occur during ordinary repairs, for example when opening the housing for a battery replacement, a lens change, or other maintenance. Without further measures, a camera according to the invention would thus be a "disposable camera" whose cryptographic operation irretrievably ends after the first opening of the housing.

To remedy this, it is proposed to provide a plurality of PUFs (per signature engine) in both the first sector and the second sector, for example five to ten discrete PUF structures per sector. During the initial initialization of the recording device, only a single PUF per signature engine is used for key formation according to Section 3.1. Its entropy contribution is combined with at least one further key component as described,

whereupon the PUF hardware is irreversibly destroyed. The private key thus formed then exists exclusively in volatile memory of the respective sector.

If this volatile memory is erased - whether through intentional triggering of the tamper protection, through a repair procedure, or through a loss of power - the recording device can be re-initialized by using, in each sector, a next, previously unused PUF for the formation of a new key pair, which is again irreversibly destroyed after key formation. Each re-initialization thus generates an entirely new cryptographic device identity. The number of PUFs provided in the recording device per sector determines the maximum number of possible re-initializations.

In principle, it would also be conceivable that, instead of a complete destruction of a discrete PUF, an irreversible physical alteration is performed on a single PUF structure, which leads to a new entropy contribution independent of the preceding one. This would be advantageous insofar as a single PUF structure could be used multiple times, or even a practically unlimited number of times. However, ensuring the independence and entropy quality of successive entropy contributions from such an altered structure is likely to be comparatively much more difficult to guarantee than when using a fresh, untouched, discrete PUF. The sequential use of a plurality of discrete PUFs is therefore the preferred embodiment.

A side effect of this approach is that a repair permitting continued cryptographic camera operation according to the invention, at least with respect to the first sector, is technically compellable only by the manufacturer: Only the manufacturer is permitted to assign the manufacturer-side key component according to Section 3.1 (this process can and should be cryptographically secured), and only the manufacturer can authenticate itself to the blockchain(s) in which the public keys of the recording device are published and the public keys of the first sector are cryptographically linked to those of the second sector (Section 7.1). Thus, from the sectored architecture and the PUF-based key management, a system-inherent manufacturer binding for repairs arises, which follows strictly technically from the security architecture. The manufacturer binding in the second sector is, by contrast, less compulsory: Since the second sector may be designed as an open-source system, and the user-side key component is assigned to the user, the manufacturer may enable the user to perform the re-initialization of the second sector with the involvement of the blockchain infrastructure.

All recordings generated prior to a re-initialization remain fully verifiable, since the associated public keys remain permanently deposited in the blockchain. After the re-initialization, the recording device merely receives a new cryptographic identity, the linkage of which to the preceding identity of the same device can likewise be deposited in the blockchain in order to make the device history traceable without gaps.

16. Implementation Notes on Suppression of Side Channels

16.1 Power Supply of the First Sector

A power supply of the first sector from the second sector is partially unavoidable, yet constitutes a breach of the otherwise very strictly sectored security topology. This appears problematic in general, but particularly so in light of the proposed call-and-response communication during lifelong self-monitoring of the camera. In this regard, it is first proposed to treat the matter as an EMC problem and to provide at least one low-pass filter which is located inaccessibly from the outside within the first sector. The following approach is preferred: Provided the first sector has an electrically conductive housing, which is very strongly preferred, this housing is connected to a common reference ground of the first and second sectors, so that only a single conductor needs to be routed into the protected proprietary sector. Preferably, the metallic housing of the first sector comprises an additional chamber, in effect a Faraday cage within the Faraday cage of the first sector. The conductor required for the power supply of the first sector may be routed into this chamber by means of a feedthrough capacitor, where it charges a buffer capacitance that functions as a low-pass filter. This buffer capacitance shall be dimensioned by a person skilled in the art, exhibit low self-inductance, and is preferably operated in parallel with a TVS diode. This buffer capacitance is effective between the housing and the conductor from the second sector. Behind this buffer capacitance there is at least one regulator, namely at least a series (linear) regulator, but preferably a switching converter with galvanic isolation of input and output, wherein the regulator shall operate autonomously, i.e., without a data connection to any other component or assembly, wherein the voltage regulation is performed exclusively on the output side, and the switching converter is preferably operated at a fixed switching frequency. The output voltage of the regulator is in turn capacitively buffered, wherein this rear buffer capacitance is, preferably, again operated in parallel with a TVS diode, and is finally routed via a further feedthrough capacitor into the "main chamber" of the first sector, in which the signature engine, image sensor, etc. may be located. Additionally, inductors may be provided to form a low-pass filter together with the aforementioned buffer capacitances. If dimensioned sufficiently large, the rear buffer capacitance functionally replaces a dedicated power supply of the first sector: Transients introduced from a compromised second sector "are lost in the noise," i.e., they can no longer be correlated with bit operations.

16.2 Optical Data Diode

If the first sector has a metallically conductive housing, this is also advantageous with respect to an optical data diode according to Claim 1: If its light guide or light beam is routed through a bore in the metallic housing of the first sector into the second sector, parasitic component capacitances within the first sector are shielded from the second sector.

16.3 Pyrofuze Wire as a Potential Side Channel

A further potential side channel exists precisely when using a Pyrofuze fuse, since this can in principle also transmit an electrical signal. A remedy may be achieved by

grounding the Pyrofuze wire. If, as described in the preceding paragraph, the housing of the first sector is electrically conductive and is used as ground, it can simply be ensured that the Pyrofuze wire, when routed into the first sector, is in reliable electrically conductive contact with the housing thereof, wherein the contact area between the Pyrofuze wire and the housing must be dimensioned such that the heat dissipation at the contact point does not inhibit the propagation of the exothermic reaction.

17. Radio Fingerprinting

17.1 VLF Fingerprinting

In the VLF fingerprinting according to the invention, it would be an error to assume that only signals of natural origin are observable in the VLF band, for example between 3 kHz and 30 kHz. On the contrary, in densely populated areas the most intense signal sources in this band may be of artificial origin, for example caused by nearby switching power supplies, frequency converters, or power electronics systems; these signals too may be used for provenance verification. The method functions, however, also without anthropogenic VLF sources, since natural sources are always and everywhere present.

A preferred exemplary embodiment of a VLF receiver is described below, which is intended to enable the person skilled in the art to carry out the invention without limiting the claimed teaching to the specifically described configuration. In particular, modifications familiar to the person skilled in the art, substitutions of individual components, or alternative dimensionings may be made without departing from the scope of protection of the invention.

A VLF receiver suitable for a mobile terminal preferably comprises a soft-magnetic rod antenna with a receiver coil. The rod antenna can be realized in a compact form factor that allows integration into common portable devices, since the spectral fingerprint sought here does not require high absolute sensitivity, but merely sufficient distinguishability of the signal sources contained in the spectrogram. A soft-magnetic ferrite rod is particularly suitable as a core, for example made of a MnZn ferrite. Instead of ferrite, laminated or powdered cores made of highly permeable nanocrystalline or amorphous soft-magnetic alloys may also be used, for example materials such as Nanoperm (Vitroperm) or Metglas. In an open magnetic circuit, as represented by a simple rod core, the effective permeability μ_{eff} is, however, limited by the demagnetization factor N: For $\mu_r >> 1/N$, μ_{eff} converges toward the value 1/N and is then nearly independent of the material permeability μ_r. Since, at length-to-diameter ratios typical in practice, the effective permeability already saturates at moderate material permeabilities, highly permeable core materials in open rod geometry offer no substantial advantage in terms of reception over MnZn ferrites with μ_r in the range of several thousand. Nevertheless, they may be expedient due to their considerably lower Barkhausen noise compared to ferrites, since the much smaller hysteresis of these materials compared to ferrites suppresses the Barkhausen noise in the core material

caused by abrupt Bloch wall displacements, which in a receiver designed for low signal levels improves the achievable noise floor. Furthermore, these materials exhibit a higher saturation magnetization and lower magnetostriction (and correspondingly lower microphonics).

For the receiver coil, a low-capacitance winding technique is preferably applied, in particular a sectional winding (bank winding), in which the winding is divided into axially separated sections. A sectional winding executed by a person skilled in the art considerably reduces the parasitic winding capacitance compared to a continuous layer winding, because the potential of adjacent turns within a section differs only slightly, while the section walls reduce the capacitive coupling between sections of different potential. The receiver coil is operated below its self-resonant frequency as a pure inductance. That is, the inductive reactance of the coil predominates in magnitude at all operating frequencies over the capacitive impedance of the parasitic winding capacitance. The number of turns is therefore preferably chosen to be high enough that the self-resonant frequency of the coil still lies above twice the highest design reception frequency. This leaves a sufficient margin to the upper band edge, so that the transfer function in the useful band remains largely frequency-independent. If the receiver coil is arranged on the same rod core as a time signal receiver coil that is tuned to a time signal frequency, for example 77.5 kHz for DCF77, then the self-resonant frequency of the VLF receiver coil shall moreover be sufficiently spaced from this time signal frequency and its harmonics, in order to avoid undesired resonance peaking and thus an overdrive of the subsequent amplifier stage.

If the arrangement is housed in a device that contains further electronic assemblies, electromagnetic shielding of the rod antenna and the preamplifier against device-internal interference sources, in particular clock oscillators and switching regulators, may be expedient.

The signal from the receiver coil is preferably fed to a low-noise differential amplifier, in particular one with low 1/f noise, for example an operational amplifier with a JFET input stage. Both terminals of the receiver coil are connected to the two inputs of the differential amplifier; the coil thereby simultaneously serves as the signal-delivering element and as a push-pull source. A common-mode signal, such as an electrostatic coupling, produces approximately the same potential at both coil terminals and is attenuated by the common-mode rejection of the differential amplifier, while the magnetically induced useful signal appears as a differential voltage between the terminals. The gain-bandwidth product of the differential amplifier is preferably dimensioned such that at the maximum signal gain a sufficient headroom remains to avoid clipping of the output signal even in the event of sporadically elevated field strengths, for example from nearby interference sources.

If necessary, the inputs of the differential amplifier may be protected against overvoltages by TVS diodes of low capacitance, for example against transients such as

those that may be induced by nearby lightning discharges. A low capacitance in the present context is to be understood as one that is smaller, and preferably several times smaller, than the stray capacitance of the receiver coil, so that the TVS diodes do not substantially affect the resonant frequency of the coil and the transfer function of the receiver.

The output signal of the differential amplifier is preferably band-limited by a low-pass filter in order to suppress signal components above the VLF useful band and to reduce the dynamic range requirements on the subsequent analog-to-digital converter. For the avoidance of aliasing, signal components with frequencies above half the sampling rate of a subsequent AD converter are in particular to be suppressed. The band limiting may be effected by means of an active low-pass filter in multiple-feedback topology (MFB filter), wherein the MFB filter preferably has an adjustable voltage gain for level adaptation.

The signal thus obtained, optionally filtered and level-adapted, is digitized by means of an analog-to-digital converter. The digitized signal is subjected to a short-time Fourier transform (STFT), wherein the temporal window length is chosen such that the spectral resolution in the VLF band is sufficient for distinguishing the signal sources present, and at the same time the temporal resolution permits assignment to a specific recording time. The resulting time-dependent spectrogram constitutes the sought VLF fingerprint and is co-signed in the second sector.

17.2 Shortwave Fingerprinting

A radio fingerprinting according to the invention is possible not only in the VLF band. A highly interesting alternative is to be seen in higher-frequency anthropogenic signals, and in particular in military, maritime, and intelligence shortwave transmitters, including number stations, and further services such as VOLMET aviation weather reports transmitted on shortwave. This is interesting not only because it is practically excluded that these very diverse sources with partly diverging interests would conspire to produce a specific signal at a specific location: They could not do so, at least not completely, because shortwave propagation depends on weather and solar activity, which are unpredictable influencing factors: Shortwave signals reach the receiver via multiple propagation paths simultaneously, namely as a ground wave as well as singly and multiply reflected "sky waves" from the ionosphere, possibly via different ionospheric layers (E, F1, F2). These partial signals travel paths of different lengths and therefore arrive at the receiver with different phase relationships. Since the ionosphere is not a static reflector, but rather its layer heights and electron densities vary constantly with time of day, solar activity, and atmospheric dynamics (weather), the relative phase relationships of the partial signals change continuously. The resulting interference alternates between constructive and destructive superposition, which is observed at the reception location as temporally and spectrally variable signal strength fluctuations.

This is known as a "fading pattern." In addition, intense and non-predictable natural and anthropogenic interference is also present in the shortwave band.

For collecting a shortwave fingerprint, a software-defined radio (SDR) is suitable, for example, which by means of an active broadband antenna observes a plurality of publicly known shortwave transmitters, wherein the signal of each of the transmitters is, e.g., digitally demodulated and subsequently subjected to a Fourier transform (STFT). Together with the temporally variable relative signal strengths, the determined time-dependent spectra constitute the fingerprint or its basis. Such fingerprinting is suitable both for recording devices according to Claim 1 and for an illumination device according to Claim 62. The described shortwave fingerprinting draws on existing, predominantly governmental, strictly classified, and highly available infrastructure on the transmitter side, without submitting to the veracity of the messages transmitted by these sources; rather, these existing transmitters can be utilized to probe the ionosphere, and this with commercially available components.

18. GNSS Protocol Supplementation

In an advantageous further development, it is proposed that navigation satellites in the future, within the framework of their civilian services, continuously co-transmit true random numbers or cryptographically strong, non-predictable pseudo-random numbers, wherein preferably each satellite transmits its own random sequence. These random numbers are received by a multitude of independent receivers worldwide and can be publicly observed, archived, and indexed by the operators of the navigation systems and/or independent third parties, so that at any point in time it can be reconstructed which random sequence was transmitted by which satellites at that point in time.

For authenticity verification, this offers a considerable advantage: Since the random numbers are by construction not predictable and their repetition, given a sufficiently large length of the random sequences, appears practically excluded under realistic constraints, a replay attack in which manipulated GNSS signals with previously transmitted random sequences are fed to a recording device would, upon subsequent verification, be revealed by the fact that the same random sequence had already been observed and indexed in the past. Any repetition of an archived random sequence at a later point in time can thus be interpreted as an indication of a signal replay.

A recording device according to the invention is configured to sign the navigation data received by its GNSS receiver in the second sector, including the respectively valid random numbers, as part of the metadata together with the recording data, further metadata of the second sector, as well as the first signature and optionally metadata of the first sector, with the private key of the second sector. Thereby, the recording is cryptographically bound to a concretely observed state of the GNSS system:

The GNSS-side random-number-based metadata thus created can advantageously be combined with external provenance protocols, in particular protocols according to the Content Provenance Profile (CPP) and/or C2PA: While the provenance protocol provides an external, publicly verifiable time anchoring and certificate chain, the GNSS random numbers signed in the camera provide an additional, physically non-predictable state anchor of the satellite system, so that replay-based GNSS spoofing attacks can be detected with high probability.

19. Attribution through Device and Person Binding

19.1 Manufacturer Attribution

Since the public key of the first sector according to Section 7.1 is published in at least one blockchain together with a sensor-individual identifier, in particular a PRNU signature of the image sensor, every recording produced with a recording device according to the invention can be cryptographically and forensically attributed to the manufacturer as well as to a specific sensor module. This also applies to recordings that were signed by means of a compromised or improperly used recording device of this manufacturer. The linkage of public key, sensor-individual identifier, and manufacturer identity creates an incentive for maintaining the physical integrity of the devices, for monitoring the manufacturing and distribution chain, and for revocation of compromised keys.

19.2 User Attribution through Identity Binding

In a preferred embodiment, the recording device is configured to perform an authentication of a user, in particular a biometric authentication, and to cryptographically include a pseudonymized identifier derived therefrom in the signature of the second sector. For this purpose, the recording device collects data for user authentication, for example using existing sensor hardware, in particular the image sensor in its capacity as an RGB-IR sensor according to Section 6, a depth sensor according to claim 33, a fingerprint sensor, and/or an iris scanner.

From the raw data, an irreversibly transformed user identifier is generated, in particular a biometric template, and in particular a pseudonymized identifier according to ISO/IEC 24745, which is renewable, revocable, and not reversible to the raw data; the raw data are deleted within the second sector after template generation and do not leave the recording device. The template is added to the data packet of the second sector prior to signing, so that the second signature attests that at the time of recording a biometrically authenticated natural person was associated with the process. If authorship is disputed, the user may have their biometrics recollected and verified against the template contained in the signed recording. The biometric authentication is optional and requires explicit activation by the user.

20. Companion Device

20.1 Basic Principle

A further aspect of the invention relates to a system comprising a recording device - hereinafter referred to as cryptographic camera - and an external companion device, which forms an additional trust domain independent of the camera. The companion device is preferably a commercially available smartphone with application software according to Section 20.8; it may alternatively be a dedicated dongle device according to Section 20.5 or comprise such a device.

The inventive basic concept consists in that the companion device collects provenance data and introduces at least a core of this provenance data, or a data set cryptographically derived therefrom, exclusively via physical signals - in particular light and sound - into the scene captured by the cryptographic camera. Thus, the companion device and the camera form a hardware data diode, and the essential principle of claim 1 is realized, only in a device consisting of two disjoint assemblies or in two separate devices instead of two separate sectors within one device.

The companion device transmits at least a part of the provenance data, or a data set cryptographically binding these, for example in the form of a two-dimensional, optically encoded pattern and/or an acoustically modulated sequence; the cryptographic camera receives these signals via its already present sensors - in particular the image sensor and/or a microphone - and signs them close to the source as part of the recording. The transmission is physically unidirectional: the companion device transmits, the camera receives; a return channel between the two does not exist. Neither a wireless coupling (Bluetooth, Wi-Fi, NFC) nor a wired connection nor an internet connection between the camera and the companion device is required.

The provenance signal contains in encoded form provenance data that preferably comprise one or more of the following items: a device identifier of the companion device, in particular a cryptographic hash value of a hardware-anchored identifier; a cryptographic authentication code that proves the integrity and authenticity of the provenance data; an irreversibly transformed user identifier, in particular a biometric template of the user of the companion device, in particular a pseudonymized identifier according to ISO/IEC 24745 of the companion device; geocoordinates, preferably GNSS-based; a time reference, preferably GNSS-based or long-wave-based or obtained from an external service; cellular network and/or radio environment data; and/or environmental radio signatures collected by means of external radio receivers.

Already the combination of a recording device with an independent companion device that introduces its own, cryptographically signed provenance data via a physically unidirectional channel into the scene captured by the camera increases the manipulation resistance, because an attacker would now have to simultaneously compromise or coordinately forge two independent devices. Preferably, the user of the

companion device is furthermore biometrically authenticated, whereby the recording is additionally bound to an identified person.

The described companion device is not limited to use with a sectored recording device according to the invention; it is equally usable with existing recording devices that have a monolithic cryptographic signature architecture, in particular with cameras that implement the C2PA standard. In this case, the camera signs the provenance signal as part of the recording without interpreting its content; the verification and association of the provenance data is performed subsequently by an external verifier. Even if the recording device does not interpret the provenance signal, the forensic value of the provenance data is preserved, since the signal is signed close to the source as part of the recording and can be subsequently extracted, decoded, and verified from the recording by an external verifier.

20.2 Optical Transmission by Means of a Display

In a preferred embodiment, the companion device displays on its screen a machine-readable optical pattern, in particular a QR code, Micro-QR code, or DataMatrix code, and is placed in the field of view of the cryptographic camera such that the screen is visible in the recording. The pattern is preferably updated periodically and contains the provenance data in full, so that it is readable from a single frame both in still image captures and in video sequences. The display may be divided into regions, in which a central region carries the code symbol and surrounding regions carry color- or brightness-modulated signals for synchronization, supplementary data, or checksums. The multi-color capability of the display can be used to increase the symbol capacity; with binary modulation of the three color channels red, green, and blue, eight combinations result, and thus three bits per area symbol; with multi-level modulation, correspondingly more.

In a variant, the display of the companion device is directed at an object in the scene, so that the pattern appears as a projection on the object and is captured by the camera; the projected pattern can simultaneously serve as an auxiliary pattern for a contrast-based and/or phase-detection-based focusing system.

20.3 Optical Transmission by Means of a Time-Modulated Light Source

In a further embodiment, the companion device transmits the provenance data by means of time-modulated light from an LED, in particular the flashlight LED of a smartphone or an LED of a dongle. The modulation is preferably performed as On-Off-Keying. The cryptographic camera receives the modulated light signal either (a) as a reflection from an object in the recorded scene, wherein the temporal modulation appears as brightness variation in the image content, or (b) by direct incidence of the LED light through the lens onto the image sensor, in particular in conditions of high ambient light or large distance, when the reflection variant yields an insufficient signal-to-noise ratio. With rolling shutter sensors, the modulated light signal can be temporally

resolved line by line; the indirect capture via the scene is preferred but not required for the implementation of the invention.

20.4 Acoustic Transmission

Additionally or alternatively, the companion device can transmit provenance data acoustically, in particular in the near-ultrasound range, via the loudspeaker of the smartphone or a piezo loudspeaker integrated in the dongle. The cryptographic camera receives the signal via an integrated or external microphone; the acoustic transmission is independent of the optical lighting conditions and is suitable as a complementary channel.

20.5 Dongle as an Extended Companion Device

In a further embodiment, the companion device comprises a dongle device connected to the smartphone via a data interface, in particular USB-OTG. The dongle preferably comprises one or more of the following components: a radio receiver for receiving long-wave time signals and/or further environmental radio signals, in particular in the form of a software-defined radio receiver; a focused LED light source with a narrower beam angle than the smartphone flash LED; a projection module for generating structured light patterns that serve as a data-carrying provenance signal and/or as a focusing auxiliary pattern; a piezo loudspeaker; a dedicated cryptographic key.

20.6 Verifying Cryptographic Camera with Conditional Image Output

In a particularly preferred aspect, the cryptographic camera is configured as a verifying camera that outputs a signed recording only when the provenance signal of the companion device has been successfully read and verified: The camera comprises a verification unit that (a) detects the provenance signal in the image and/or audio signal, (b) decodes the provenance data, (c) checks the cryptographic authentication code and preferably performs one or more plausibility comparisons (comparison of the received timestamp with the camera-internal timestamp, comparison of the received geocoordinates with the camera-internal geocoordinates, checking of a user identifier and/or a device identifier against an authorization list), and (d) upon successful verification outputs the recording as a signed file.

Upon failed verification, the recording is discarded and not persisted, or stored unsigned with a machine-readable note regarding the failed verification. The authorization list can be organized on a person-based, device-based, or role-based basis and stored locally in the camera or updated via a secure data connection. It is strongly preferred that the detection and decoding of the provenance signal takes place only after the image has already been signed on the camera side, whereby the process of signing can be separated in hardware from the interpretation of the image material. At this point, multi-factor authorization is possible and provides additional protection: The camera itself, or otherwise independently of the camera at a later time, can contact the authenticated

user of the companion device, for example by e-mail, SMS, messenger service, or dedicated application software, and request confirmation of their participation in the signed recording. Additionally, C2PA-capable recording devices themselves can also be configured to offer, or to require, user legitimation, in particular biometric legitimation.

20.6a Trust Level

The recording device or a verifier can assign to each recording a trust level that reflects the number and type of provenance data delivered and verified by the companion device (device identity, geocoordinate-time consistency, biometric authentication, radio environment, radio fingerprint). An image or video output occurs only when the trust level reaches or exceeds a configurable minimum; for video, it can be built up cumulatively over the recording duration. If it falls below the minimum, affected segments are marked or blocked.

20.7 Application to Regulated Supply Chains

A preferred field of application is the documentation of critical tracking events in regulated supply chains, in particular in the food sector. In this case, a documentation officer authenticates at the companion device, in particular biometrically, which thereafter generates an optical and/or acoustic provenance signal with the provenance data according to Section 20.1 and is placed visibly in the recording field of the cryptographic camera or directed at an object to be documented.

The verifying camera according to Section 20.6 captures the scene, verifies the provenance signal, and upon successful verification outputs a signed recording whose provenance manifest contains both image metadata and the verified provenance data of the companion device. Each documented station of the supply chain is thereby cryptographically bound to the manufacturer of the camera, to the recording device, to the time and place of the recording, as well as to an authenticated and, where applicable, biometrically associated person.

20.8 Application Software on a Smartphone

20.8a Collection of Connection Metadata by the Companion Device

The companion device, in particular a smartphone, preferably has a cellular network module and/or another network interface and thus the capability to establish data connections to external services during provenance collection. Each such connection leaves cryptographically signed traces at the contacted service that can neither be generated nor controlled in content by the companion device and that are subsequently independently verifiable by a verifier. The companion device deliberately exploits this property by establishing at least one data connection to an external service at the time of provenance collection and collecting the resulting connection metadata signed by third parties as provenance data and incorporating them into the provenance manifest.

The established data connections relate exclusively to the communication between the companion device and external services and not to the transmission channel to the camera, which according to Section 20.1 occurs exclusively via physical signals. The thus recruited third parties - in particular certification authorities, Certificate Transparency Log operators, and Time Stamping Authorities - are not knowingly involved in the provenance attestation; they merely provide their regular services. The unawareness of the recruited third parties makes collusion with the companion device unlikely and lends the data signed by them a forensic value that exceeds that of device-own data.

The collected connection metadata comprise in particular, individually or in combination:

(a) an OCSP Stapled Response according to RFC 6960,

(b) one or more Signed Certificate Timestamps (SCTs) according to RFC 6962 or RFC 9162,

(c) a response of a Time Stamping Authority (TSA) according to RFC 3161,

(d) the complete TLS certificate chain of the contacted server,

(e) DNS resolution data including, where applicable, DNSSEC signatures.

20.8b Collection of Cellular Network and Radio Environment Data by the Companion Device

The companion device collects locally available cellular network and radio environment data that enable an independent location determination and cross-validate the GNSS data. These comprise in particular: (a) cell identifiers of the serving and neighboring cellular network cells (Cell ID, MCC, MNC, LAC or TAC together with signal strengths); (b) propagation delay parameters (e.g., Timing Advance value); (c) MAC addresses of visible Wi-Fi access points with received signal strengths.

20.8c Verification of the Provenance Data of the Companion Device

The recording device or a downstream verifier can decode the machine-readable provenance signal contained in the recorded signal and verify the provenance data against sensor data or reference data. In particular, geocoordinates, time references, radio environment data, and the signature of the companion device are compared. The result of the verification can be incorporated as a metadata item into a signature of the recording device or a provenance manifest.

20.8d Offline Operation

The system according to Section 20.1 is preferably also functional without an existing internet connection of the companion device. In this case, the connection metadata according to Section 20.8a are omitted; the remaining provenance data - in particular a

device identifier of the companion device, preferably secured by a strong authentication method, in particular a biometric one, as well as, where available, GNSS data and/or cellular network and radio environment data according to Section 20.8b - are collected independently of an internet connection and incorporated into the provenance signal.

For offline verification, a public key of the companion device is deposited in the recording device or in a verification infrastructure. The companion device signs the signal with its private key; verification is performed against the deposited public key without a network connection. After restoration of an internet connection, the companion device can retrospectively enrich the data collected in the interim with connection metadata and add a signed timestamp from a TSA.

20.8e Data Budget and Prioritization of the Provenance Signal

The machine-readable provenance signal is divided into a mandatory part and a supplementary part. The mandatory part comprises at least a device identifier of the companion device and a signature of an asymmetric signature scheme with compact signature length, for example Ed25519, generated over all other fields of the mandatory part, wherein the private key is held in a secure hardware element and is preferably released only after successful user authentication, in particular biometric authentication.

The supplementary part contains further provenance data (cellular network / radio environment, connection metadata, etc.) and is bound to the mandatory part via a cryptographic hash value.

20.9 Extensions of Cryptographic Provenance Protocols

A further aspect relates to extensions of cryptographic provenance protocols, in particular the C2PA standard and the CAWG Identity Assertion standard, to support provenance attestation by an external companion device. The extensions comprise in particular: (a) a user-defined assertion under an application-specific namespace in the provenance manifest, which contains the received and verified provenance data in structured form; (b) an extension of the signing logic of the cryptographic camera that performs the verification according to Section 20.6 prior to signing and includes the verification status in machine-readable form in the assertion; (c) an optional Identity Assertion, in which the user of the companion device is included as an additional actor with a role in the manifest; (d) an extension of the Trust List logic for managing the authorization list according to Section 20.6; and (e) a content binding, in which the manifest contains a hash of the optical pattern contained in the image, so that the link between the visible pattern and the extracted provenance data is cryptographically verifiable.

21. Device-External Challenge with Time-Critical Object Illumination as Proof of Authenticity

In a further embodiment, which is also executable independently of the sectored architecture according to claim 1, a method for impeding real-time forgery of audiovisual recordings by means of generative artificial intelligence is described. The generation of a photorealistic, physically plausible video image by a generative system typically requires a non-negligible computation time, in particular when the video image is to correctly reproduce a temporally varying, spatially structured illumination whose specific course was not known in advance. If an attacker is granted only a short, device-externally specified and previously unknown time window within which an optically consistent response must be produced, the effort required for a deceptive forgery increases considerably.

For this purpose, an illumination means (e.g., of a second sector in recording devices according to claim 1) or companion device cooperates with a device-external entity. The device-external entity lies outside the sphere of influence of the camera user and can in particular be a cellular network service, a service provided by the mobile network operator or its infrastructure, a network-proximate edge service, a service of a communication or conferencing platform, a time stamping entity, a Transparency Log service, or another network service independent of the recording device. This entity generates a challenge that comprises at least one parameter for defining a temporally varying illumination pattern and thereby specifies the beginning of a time window within which the illumination pattern is to be executed and the optical response is to be captured. The challenge is preferably digitally signed and provided with a network-internal timestamp, or its cryptographic hash value is deposited with a time anchoring service, so that origin and time of generation are provable after the fact.

The illumination means receives the challenge via a communication interface and derives therefrom a temporally varying and preferably spatially structured illumination pattern with which an object to be recorded and/or a scene is irradiated. The illumination is performed such that the illumination pattern is detectable in the image or video data recorded by a camera as a physical response, in particular in the form of temporally varying brightness, color, shadow, reflection, interference, and/or structure patterns. In a preferred embodiment, the illumination device comprises at least one directed light source, in particular a laser diode, a VCSEL source, or a VCSEL array. The light source may comprise an optical shaping element, in particular a diffractive optical element, a diffraction grating, a diffuser, a micro-optics array, a projection module, or a combination thereof, so that the beam can be split into a plurality of separate reflections, spots, or structures. This makes it possible, on the one hand, to generate a complex illumination pattern and, on the other hand, to distribute the intensity across multiple partial beams, so that applicable eye safety requirements can be met.

In a preferred embodiment, coherent or partially coherent radiation is used, so that a speckle pattern arises on the illuminated object surface, which depends on the microscopic surface structure and can be evaluated as a further component of the

optical response pattern. The speckle pattern is, in embodiments with proof of authenticity, not suppressed but rather used as an additional physical signature resulting from the interaction of illumination and object.

A recording device, in particular a camera with cryptographic signing of the recording data, captures during illumination the optical feedback of the illumination pattern and generates image or video data that contain the optical response pattern. The recording device can be a recording device according to claim 1, a camera with C2PA support, or another cryptographically signing recording device; the present embodiment is, however, not limited thereto. Preferably, the recording data are indeed cryptographically signed in the recording device, so that a provenance chain arises; however, the temporal classification of the optical response pattern that is decisive for the proof of authenticity is not derived solely from camera-internal time data but is determined by a device-external entity on the basis of external timestamps or reception times.

In a preferred embodiment, the camera has an image sensor with rolling-shutter-induced temporal staggering of the exposure, in which different sensor rows or columns have different effective exposure times. This creates within a nominal single frame an additional temporal structure: A temporally varying illumination pattern leaves a row-by-row or column-by-column varying brightness and/or structure progression that can be physically consistent only if the illumination was actually present at the respective effective exposure time of the relevant sensor row or column. A forging entity that seeks to replicate an optical response pattern must therefore not only correctly reproduce the temporal progression specified by the challenge across a sequence of individual frames, but additionally, within each individual frame, consistently render the rolling-shutter-induced temporal staggering, which further increases the effort required for a real-time forgery.

In a subsequent or ongoing authenticity check, it is verified whether the recorded image or video data exhibit an optical response pattern consistent with the challenge. For this purpose, the pattern observable in the image material is correlated with the temporal and, where applicable, spatial progression of the illumination pattern expected from the challenge. A statistically significant match within the relevant time window proves that the recording was created under the specified illumination conditions and within the required time. If the match is absent, lies outside the permissible time window, or the pattern exhibits physically implausible artifacts, such as inconsistent shadow casting, interreflections, material responses, or interference patterns (in particular speckle patterns), this constitutes an indication of manipulation or forgery by a generative system.

Essential for the evidentiary strength is that the check of whether the challenge was passed is performed entirely or partially device-externally and thus outside the sphere of influence of the camera user (and preferably also that of the camera manufacturer). The device-external entity or an external verification entity cooperating therewith can in

particular generate a cryptographically secured verification protocol that documents the passing or non-passing of the challenge.

21.1 Time Binding Without Trust in the Camera

In a preferred embodiment, the device-external entity is a network service, in particular a mobile network operator at the level of a base station, a core network node, or an edge service arranged at the network edge, which both generates and signs the challenge with its own time base and also provides the video stream transmitted by the camera, or a cryptographic hash value derived therefrom, with a likewise network-internally signed timestamp, so that an authenticity check can be performed without having to trust the camera-internal time base.

In dimensioning the permissible reaction time window between transmission of the challenge and reception of the optical response, the sum of the involved physical and processing latencies forms a mandatory lower bound; these include in particular the device-internal latency for image capture, processing, and signature generation, as well as the transmission delay between camera and network service, including the latencies in the access network and, where applicable, in the core network. In order to prevent a time-consuming image synthesis by generative systems, the permissible response time is at the same time limited upward to a maximum closely aligned with this lower bound. A person skilled in the art takes into account known or conservatively estimated timing parameters and allows for a minimal but sufficient tolerance window.

A separate time stamping service on the public internet is not required for this purpose but can additionally be used to support long-term, network-independent archiving of the hash values.

21.2 Use of Cellular Network Pseudo-Random Sequences

In a further embodiment, the parameter of the challenge is wholly or partially derived from pseudo-random sequences defined in cellular network standards, for example from scrambling or reference signal sequences of 5G-NR, which are generated by means of standardized pseudo-random generators, in particular Gold sequences with standardized initialization. Since these sequences are not predictable for an outsider without knowledge of the respective cell and time parameters, yet are reconstructable by a verifier from publicly documented 3GPP specifications, they are suitable as an external source, independent of the recording device, of deterministic but, for an attacker, not pre-computable control data for the temporally varying illumination pattern.

The foregoing embodiment is not limited to cellular networks or to a specific standard. What is decisive is that a device-external entity derives the challenge from a sequence that is not pre-computable for the attacker yet is reconstructable by a verifier, that the decisive time binding takes place outside the sphere of influence of the camera user,

and that the check of the passing of the challenge is performed entirely or partially device-externally.

CLAIMS

Group A: Sectored Architecture and Data Diode

Independent Claim 1 - Sectored Recording Device

1. A recording device for generating tamper-resistant data, comprising:

a) a first sector, which comprises at least one sensor and a first signature engine, wherein the first signature engine is configured to determine cryptographic hash values from a signal captured by the sensor, and to digitally sign these with a private key of a key pair assigned to the first sector, in order to generate a first signature;

b) a second sector, which comprises at least a second signature engine, wherein the second signature engine is configured to receive the signal and the first signature and to generate a second signature using a private key of a key pair assigned to the second sector;

c) a data diode arranged between the first sector and the second sector, which effects a physically enforced unidirectional data transmission from the first sector to the second sector and prevents an information backflow from the second sector to the first sector;

wherein the first sector and the second sector are assigned to different trust domains.

1a. A device for generating tamper-resistant data, comprising:

a) at least one first sector, which comprises at least one data source and a first signature engine, wherein the first signature engine is configured to determine cryptographic hash values from data provided by the data source and to digitally sign these with a private key of a key pair assigned to the first sector, in order to generate a first signature;

b) at least one second sector, which comprises at least a second signature engine, wherein the second signature engine is configured to receive the data and the first signature and to generate a second signature using a private key of a key pair assigned to the second sector;

c) at least one data diode arranged between a first sector and a second sector, which effects a hardware-enforced unidirectional data transmission from the first sector to the second sector and prevents an information-carrying backflow of digital data from the second sector to the first sector;

wherein at least two of the sectors are assigned to different trust domains;

and wherein between the sectors, in addition to the data diode, a power supply path for energy supply is permissible, provided that the power supply path is configured by

circuit-engineering measures such that an information transmission over the power supply path suitable for reconstruction or substantive influencing of data generated or processed in a sector is suppressed, so that the power supply path does not negate the informational isolation effected by the data diode.

1b. The device according to claim 1a, wherein the circuit-engineering measures for suppressing a data-carrying modulation of the power supply path within the supplied sector comprise at least one of:

a) a buffer capacitance, the dimensioning of which prevents a correlation between input-side voltage fluctuations and bit-synchronous operations in the supplied sector;

b) a voltage regulator, in particular a switching regulator with galvanic isolation of input and output, which operates autonomously and without a data connection to other assemblies;

c) a low-pass filter with a cutoff frequency that is substantially below the lowest clock frequency of the supplied sector;

d) a TVS diode for protection against overvoltage transients.

1c. The device according to claim 1a or 1b, wherein the power supply path into the supplied sector is routed via a first feedthrough capacitor into an electromagnetically shielded chamber within the supplied sector, wherein at least one autonomously operating voltage regulator is arranged in the shielded chamber, the output voltage of which is conducted via a second feedthrough capacitor into a main chamber of the supplied sector, so that input-side transients at the output of the voltage regulator are no longer correlatable with operations in the main chamber.

1d. The device according to any one of claims 1a to 1c, wherein the power supply path additionally serves as a control path for a predetermined, finite set of non-security-relevant operating commands, in particular power on/off and/or mode selection, provided that:

a) the set of permissible commands is limited by hardware to a predetermined number of discrete states;

b) no command can effect a change of firmware, cryptographic material, or memory contents in the receiving sector; and

c) the command transmission takes place exclusively via the temporal duration of a voltage state and is hardened against high-frequency modulation by the measures according to claim 1b or 1c.

1e. The device according to any one of claims 1a to 1d, comprising at least three sectors and at least two data diodes, wherein at least three of the sectors are assigned to pairwise different trust domains, so that a directed, cross-sector chain of trust is formed.

1f. The device according to any one of claims 1a to 1e, wherein the data diode is realized by at least one of:

a) an optoelectronic coupler;

b) a fiber-optic connection with exclusively one light-transmitting and one light-receiving unit;

c) a circuit with strongly asymmetric transmission characteristics, which physically prevents signal transmission in the reverse direction or attenuates it to such an extent that an information backflow is prevented;

d) a capacitive or inductive isolator with unidirectional transmission characteristics;

e) a formally verified, unidirectional hardware gate in an application-specific integrated circuit or field-programmable gate array, wherein the formal verification demonstrates the absence of a return channel.

1g. The device according to any one of claims 1a to 1f, wherein at least one sector comprises a sensor, in particular selected from: image sensor, microphone, chemical sensor, physical sensor, medical sensor, industrial transducer, biosensor, inertial sensor, and/or IoT sensor, and wherein the signature engine of the sector comprising the sensor is configured to determine and sign cryptographic hash values from a signal captured by the sensor close to the source.

1h. The recording device according to claim 1, which further comprises all features of any one of claims 1a to 1g.

2. The recording device according to claim 1, wherein the first sector is assigned to the manufacturer of the recording device and the second sector is assigned to the user of the recording device.

3. The recording device according to any one of the preceding claims, wherein the sensor comprises an image sensor and the signal is an audiovisual signal.

4. The recording device according to any one of the preceding claims, wherein the data diode is realized by at least one of:

a) an optoelectronic coupler, in which a light-emitting component in the first sector drives a light-sensitive component in the second sector;

b) a fiber-optic connection with exclusively one light-emitting unit and one light-receiving unit;

c) a circuit with strongly asymmetric transmission characteristics, which physically prevents signal transmission in the reverse direction or attenuates it to such an extent that an information backflow is prevented.

5. The recording device according to any one of the preceding claims, wherein the first sector is arranged wholly or partially within the second sector.

6. The recording device according to any one of the preceding claims, wherein the first sector is configured in a manufacturer-specific and proprietary manner and the second sector is configured as an open-source system whose hardware and/or software specification is publicly viewable and verifiable.

7. The recording device according to any one of the preceding claims, wherein the first signature engine is configured to include, in addition to the data captured by the sensor, also metadata generated in the first sector – in particular autofocus information, depth sensor data, and/or liveness parameters – in the hash value computation together with the sensor data and to sign them before leaving the first sector.

8. The recording device according to any one of the preceding claims, wherein the first signature engine is integrated as a System-on-Chip (SoC) on the chip of the image sensor, so that the signal path between data source and signing point is minimized.

9. The recording device according to any one of the preceding claims, wherein in the first sector a RAW data stream and at least one compressed data stream are generated in parallel and both data streams are signed independently of each other before being communicated to the second sector via the data diode.

10. The recording device according to any one of the preceding claims, wherein the second sector further comprises a plurality of sensors, selected from: GNSS module, long-wave time signal receiver, inertial sensors including gyro sensors and acceleration sensors, environmental sensors, and wherein the metadata captured by these sensors are added to the signal before the second signing.

11. The recording device according to any one of the preceding claims, wherein the recording device is configured to activate and deactivate the first sector from the second sector by controlling the supply voltage, wherein it is ensured by hardware that no changes to a firmware of the first sector can be effected thereby.

12. The recording device according to claim 11, wherein the firmware of the first sector distinguishes between at least two operating modes based on the duration of the applied supply voltage:

a) a challenge-response mode, in which, upon a voltage pulse with a duration below a predetermined threshold duration, a single signed image is generated and output via the data diode; and

b) a regular recording mode, in which, when a supply voltage is applied beyond the threshold duration, continuous recording operation is commenced.

13. The recording device according to claim 12, wherein the second sector monitors the response time of the first sector to a voltage pulse and evaluates an absence of the expected response or an impermissible deviation of the response time from a time profile characteristic of the first sector as a tampering indicator.

Group B: Key Generation and PUF Destruction

Independent Claim 14 - PUF-Based Key Generation with Irreversible PUF Destruction

14. A method for key generation for a cryptographic system, comprising the steps of:

a) obtaining a hardware-bound entropy contribution by means of a Physical Unclonable Function (PUF), which utilizes manufacturing-induced variations of a semiconductor structure;

b) combining the entropy contribution with at least one further key component to generate a private key;

c) storing the private key in a volatile memory;

d) after completion of the key generation, irreversibly destroying the PUF hardware, so that a renewed obtaining of the entropy contribution is physically impossible;

wherein after step d) the private key exists exclusively in the volatile memory.

15. The method according to claim 14, wherein the irreversible destruction of the PUF hardware is effected by applying a voltage to at least one MOS device of the PUF that produces an irreversible gate oxide breakdown.

16. The method according to claim 14 or 15, wherein the private key is composed of at least three components: the PUF entropy contribution, a partial information introduced by the manufacturer, and a partial information introduced by the user, so that neither the manufacturer, nor the user, nor the PUF hardware alone knows or can reconstruct the complete private key.

17. The method according to any one of claims 14 to 16, wherein the volatile memory is an SRAM memory that is permanently powered by a battery, and wherein a voltage loss leads to the immediate and irreversible loss of the private key.

18. The recording device according to any one of claims 1 to 13, wherein in at least one of the sectors a private key is generated and stored according to the method of any one of claims 14 to 17, and wherein preferably in each of the sectors a respective private key is generated and the respective PUF is irreversibly destroyed, so that the private keys after initialization exist exclusively in volatile memories in at least two disjoint sectors.

Group C: Gas Pressure Monitoring

Independent Claim 19 - Gas Pressure Monitoring for Tampering Detection

19. An electronic device with tamper protection, comprising:

a) at least one memory in which security-relevant data are held;

b) at least one gas-tight housing in which the memory is arranged and which is filled with a gas whose pressure deviates from an ambient pressure;

c) at least one pressure sensor and at least one temperature sensor, which measure the pressure and the temperature of the gas quasi-continuously;

d) an evaluation unit which is configured to trigger an erasure of the security-relevant data as soon as at least one of the following conditions is detected:

- i) a pressure change rate that exceeds a predetermined threshold value;
- ii) an absolute pressure outside a permissible pressure range;
- iii) a deviation between the measured pressure and a pressure expected on the basis of the measured temperature and a known volume according to the ideal gas law, which exceeds a predetermined tolerance range;
- iv) a temperature below a predetermined minimum value, in particular for protection against cold boot attacks.

20. The electronic device according to claim 19, wherein the gas is an inert gas and/or wherein the chemical composition of the gas fill is monitored by means of electrochemical or optical sensors, wherein the exact composition is a manufacturer secret.

21. The electronic device according to claim 19 or 20, wherein the evaluation unit is further configured to initiate, upon detected tampering, in addition to the data erasure, a thermal destruction of the memory, in particular by short-circuiting a battery powering the memory.

22. The electronic device according to any one of claims 19 to 21, wherein upon loss of housing integrity, a chemical destruction of the memory is provided by reaction of memory components or a reactive substance surrounding them with constituents of air.

23. The recording device according to any one of claims 1 to 13 and 18, wherein at least one sector is configured as an electronic device according to any one of claims 19 to 22.

Group D: Cross-Sector Key Erasure

Independent Claim 24 - Key Erasure by Diffusion-Controlled Chemical Reaction

24. An electronic device having at least two spatially separated security zones, wherein:

a) in each security zone at least one volatile memory is arranged that contains security-relevant data and is powered by an associated battery, wherein each battery is associated with a fuse element;

b) the fuse elements of the different security zones are thermally coupled by at least one continuous reactive element, wherein the reactive element has a first section in a first security zone and a second section in a second security zone;

c) a triggering of the fuse element in one security zone initiates an exothermic, self-sustaining chemical reaction of the reactive element, which propagates along the reactive element and inevitably also triggers the fuse element of the other security zone;

so that a key erasure command crosses a security zone boundary as a diffusion-controlled chemical reaction without an electrical or electromagnetic signal being transmitted.

25. The electronic device according to claim 24, wherein the reactive element comprises a wire made of an aluminum-palladium alloy (Pyrofuze) or a reactive multilayer film based on aluminum and nickel (NanoFoil).

26. The electronic device according to claim 24 or 25, wherein the reactive element additionally serves as a fuse element of one or more of the batteries.

27. The recording device according to any one of claims 1 to 13, 18, and 23, wherein the first and the second sector are thermally coupled by a reactive element according to any one of claims 24 to 26.

Group E: Permanent Cryptographic Self-Observation

Independent Claim 28 - Cryptographic Image Chain

28. A recording device for generating tamper-resistant recording data, comprising at least one sensor for capturing image and/or audio data and at least one signature engine,

characterized in that the recording device is configured to generate, throughout its entire operating lifetime – including outside of regular recording operation – compressed background images by means of the sensor at time intervals whose expected value lies in the range of seconds to a few minutes, wherein:

a) each background image is signed close to the source in a first security zone and communicated to a second zone, in which it is subjected to a second signing together with current metadata;

b) each chain element is cryptographically chained to the preceding chain element, in that a cryptographic hash value of the preceding chain element is included in the signature or hash value computation of the subsequent chain element; and

c) the resulting image chain is stored locally in encrypted form;

so that a seamless, tamper-proof image chain over the entire operating lifetime of the recording device is created.

28a. The recording device according to any one of claims 1 to 13, wherein the recording device is configured to generate, throughout its entire operating lifetime – including outside of regular recording operation – compressed background images by means of the sensor of the first sector at time intervals whose expected value preferably lies in the range of seconds to a few minutes, wherein:

a) each background image is signed close to the source in the first sector, unidirectionally communicated to the second sector via the data diode, and there subjected to a second signing together with current metadata;

b) each chain element is cryptographically chained to the preceding one, in that the cryptographic hash value of the preceding chain element is included in the subsequent one;

c) the resulting image chain is stored locally in encrypted form in the second sector;

so that a seamless, tamper-proof image chain over the entire operating lifetime of the recording device is created.

29. The recording device according to claim 28, wherein the time intervals between the background images are determined pseudo-randomly.

30. The recording device according to claim 28 or 29, wherein a Merkle tree is formed over the chain elements, which enables a verifiable disclosure of a temporal segment of the image chain without having to disclose the entire chain.

31. The recording device according to any one of claims 28 to 30, wherein the image chain further comprises the challenge-response frames generated according to claim 12.

32. The recording device according to any one of claims 28 to 31, wherein the image chain is stored symmetrically encrypted, wherein the symmetric key is known exclusively to the user.

Group F: Anti-Screen-Recording and Liveness

33. The recording device according to any one of claims 1 to 13, wherein the first sector further comprises at least one of the following systems for detecting screen reproductions:

a) a depth sensor, wherein preferably an existing autofocus system is incorporated;

b) a system with active object illumination, in particular in the near-infrared range;

c) an ultrasonic interaction system for determining three-dimensional object properties;

d) a liveness sensor, in particular an image sensor configured as an RGB-IR sensor (4-channel sensor), which simultaneously captures visible light and infrared radiation;

wherein signals of these sensors are signed close to the source in the first sector.

34. The recording device according to claim 33, wherein imaging sensors are assigned to the first sector and illumination means are assigned to the second sector, so that a compromise of the second sector does not impair the authenticity of the imaging sensor system.

Group G: Electromagnetic and Chemical Protection

Independent Claim 35 - Electromagnetic and Chemical Protection of a Cryptographic Assembly

35. An electronic assembly with cryptographic function, comprising:

a) a cryptographic circuit, which is embedded in a potting compound, wherein the potting compound is selected such that:

- i) it is insoluble in organic and halogenated solvents at standard conditions; and
- ii) all solvents, including supercritical fluids, that can substantially swell or dissolve the potting compound also attack and functionally destroy the potted electronics;

b) an electromagnetic shielding, which at least partially surrounds the cryptographic circuit, wherein for optical feedthroughs an electrically conductive, optically transparent coating, in particular an ITO coating, is provided;

c) a consistently differential signal transmission arranged within the shielding, at least outside of integrated circuits, for suppression of common-mode interference.

36. The electronic assembly according to claim 35, wherein the potting compound contains fillers for influencing its thermal conductivity and/or electromagnetic properties, in particular aluminum nitride (AlN) and/or ferrite powder.

37. The electronic assembly according to claim 35 or 36, wherein the electrically conductive, optically transparent coating is applied to the rear lens of an objective and forms, in conjunction with a conductive housing, a substantially closed Faraday cage, wherein the coating is connected to the housing via a permanent seal.

38. The recording device according to any one of claims 1 to 13, wherein at least the first sector is configured as an electronic assembly according to any one of claims 35 to 37.

Group H: Data Diode Protection and Signal Path Monitoring

39. The recording device according to any one of claims 1 to 13, wherein at least one tamper protection device of one of the sectors is configured to also detect tampering attempts that affect a transmission path of the data diode that lies outside both sectors.

40. The recording device according to any one of claims 1 to 13, wherein at least in the second sector a key erasure is triggered as soon as an expected signal of the data diode is absent, wherein the expected signals preferably follow one another at pseudo-random intervals.

41. The recording device according to any one of claims 1 to 13, wherein at least one signal path between the sensor and the first signature engine within the first sector is monitored and a change of this signal path is evaluated as a tampering indicator.

42. The recording device according to any one of claims 1 to 13, wherein upon detection of an electromagnetic attack, in particular an overvoltage event, an immediate key erasure is triggered.

Group I: PQC and Blockchain

43. The recording device according to any one of claims 1 to 13, wherein at least one of the signature engines uses a post-quantum cryptography algorithm (PQC algorithm).

44. The recording device according to claim 43, wherein the first and the second signature engine use different PQC algorithms that are based on different mathematical problems, in particular selected from: lattice-based problems, code-based problems, hash-based schemes, and/or multivariate polynomial equation systems.

45. The recording device according to any one of claims 1 to 13, wherein the public keys of the first and the second sector are deposited in a linked manner in a plurality of mutually independent blockchains, so that an attacker would have to simultaneously overcome the independent consensus mechanisms of a plurality of blockchains.

Group J: Atmospheric Signals

46. A system for authenticity verification of a data set, comprising:

a) at least one sensor for capturing an electromagnetic environmental signal in a frequency range that is characteristic of atmospheric and/or ionospheric propagation conditions, in particular in the VLF range (Very Low Frequency) or in the shortwave range;

b) a processing unit which is configured to:

- i) determine from the captured electromagnetic environmental signal at least one characteristic parameter or a feature pattern, in particular characteristic

quantities of sferics, tweeks, and/or VLF hiss, and/or spectral power densities, and/or time-dependent spectrograms; and

- ii) associate the parameter or the feature pattern temporally and spatially with a recording event;

c) a comparison unit which is configured to:

- i) provide reference data on electromagnetic environmental signals, wherein the reference data were captured and aggregated by at least one reference receiver and/or by an external observation infrastructure; and
- ii) compare the determined parameter or the feature pattern with the reference data and derive from the comparison a measure for the plausibility of the claimed location and/or time of the recording event;

wherein the system is configured to make, on the basis of the measure, a statement about the authenticity of the data set with respect to its location and/or time information.

47. The system according to claim 46, wherein the reference receiver is a further recording device, in particular a recording device according to any one of claims 1 to 13.

48. The system according to claim 46 or 47, wherein the external observation infrastructure comprises at least one of an earth observation satellite, a network of geophysical sensors, or a lightning location network.

Group K: Platform System for Authenticity Verification

Independent Claim 49 - Platform System

49. A data processing system, in particular a video platform, social media platform, a web browser, or a browser application, which is configured to:

a) check, for a received audiovisual data element, whether it can be assigned through a flawless certificate chain to a recording with a recording device that has a multi-stage signature chain based on at least two independent trust domains;

b) signal to a user whether the audiovisual data element has been verified as authentic; and/or

c) enable the user to activate a filter in which preferably exclusively recordings verified as authentic are displayed.

50. The data processing system according to claim 49, wherein the check according to feature a) comprises verifying a first signature of a proprietary sector and a second signature of an open sector based on cryptographically linked public keys

deposited in at least one blockchain, and wherein the verification of both signatures is required for a positive authenticity assessment.

51. A method for external time anchoring of recording data of a recording device, comprising the steps of:

a) generating cryptographic hash values from a signal captured by the recording device;

b) transmitting the cryptographic hash values to a plurality of mutually independent device-external time anchoring services;

c) receiving time anchoring receipts from the services;

wherein the selection of the specific services from a service list is performed by a method whose result is subsequently verifiable but not predictable in advance.

52. The method according to claim 51, wherein the plurality of services belong to different categories, in particular Time Stamping Authorities (TSA), publicly auditable transparency logs, and blockchains.

53. The method according to claim 51 or 52, wherein the selection of the specific services is derived by applying a cryptographic hash function to an unpredictable environmental signal captured by the recording device, in particular an atmospheric electromagnetic signal in the VLF range, in conjunction with a publicly known service list.

54. The method according to any one of claims 51 to 53, wherein the recording device locally generates and buffers the cryptographic hash values and associated provenance data, and the transmission to the external services takes place only when a network connection exists or is established,

wherein device-local time references establish a lower time bound and the external time anchoring establishes an upper time bound for the time of formation of the hash values.

55. The method according to any one of claims 51 to 54, wherein the recording device independently captures connection metadata of the communication with the external services, in particular its own network address, TLS certificate chains of contacted services, DNS resolution data, and/or results of a traceroute performed by the recording device itself, and includes these connection metadata in the cryptographic hash value computation.

56. The recording device according to any one of claims 1 to 13, which is further configured to perform a method according to any one of claims 51 to 55, wherein the transmission to the external services and the processing of received time anchoring receipts take place outside the first sector.

57. A cryptographic system, comprising:

a) N Physical Unclonable Functions (PUFs), wherein $N \ge 2$;

b) a volatile key memory;

c) a control unit which is configured, upon an initialization or re-initialization of the system, in each case to:

- i) select a next, hitherto unused PUF;
- ii) obtain an entropy contribution from the selected PUF and combine it with at least one further key component to form a private key;
- iii) store the private key in the volatile key memory; and
- iv) irreversibly destroy or irreversibly physically alter the selected PUF;

wherein N determines the maximum number of possible initializations and each initialization generates an identity that is cryptographically independent of all preceding initializations.

58. The cryptographic system according to claim 57, wherein upon loss of the content of the volatile key memory, in particular by triggering of a tamper protection device, by a repair procedure, or by a voltage loss, a re-initialization according to feature c) with a next unused PUF is performable.

59. The cryptographic system according to claim 57 or 58, further comprising an interface to at least one blockchain, wherein:

a) upon each initialization, a public key associated with the private key is deposited in the blockchain; and

b) upon a re-initialization, the new cryptographic identity is linked with the preceding identity of the same system in the blockchain;

so that all signatures generated before a re-initialization remain verifiable based on the public keys permanently deposited in the blockchain and the device history is seamlessly traceable.

60. The recording device according to any one of claims 1 to 13 and 18, wherein in at least one sector a cryptographic system according to any one of claims 57 to 59 is arranged, wherein preferably in each sector a plurality of PUFs per signature engine is provided, and wherein a re-initialization of the first sector requires the introduction of a manufacturer-side key component.

61. The recording device according to claim 34, wherein at least one illumination means of the second sector is configured to illuminate the object to be recorded with modulated light carrying data provided by the second sector, in particular geolocalization data, time information, and/or an environmental fingerprint, in particular a radio environmental fingerprint, and in particular a VLF fingerprint, wherein the image sensor of the first sector captures the modulated illumination as part of the recorded signal, which is subsequently signed in the first sector.

62. An illumination device for data-carrying object illumination, configured for use with a recording device having cryptographic signing, wherein the illumination device can illuminate an object to be recorded with modulated light carrying provenance data, in particular geolocalization data, time information, and/or an environmental fingerprint, in particular a radio environmental fingerprint, and in particular a VLF fingerprint.

63. A recording device for generating tamper-resistant recording data, comprising

a) at least one sensor for capturing image and/or audio data;

b) a GNSS receiver which is configured to receive navigation data of a satellite navigation system, wherein the navigation data comprise at least one sequence of true random numbers or cryptographically strong, unpredictable pseudo-random numbers provided by the satellite navigation system;

c) a signature engine which is configured to:

- i) form a data set package from the recording data captured by the sensor and metadata,
- ii) include the navigation data received by the GNSS receiver, including the respectively valid random numbers, as further metadata in the data set package, and
- iii) compute cryptographic hash values over the data set package and sign them with a private key assigned to the recording device,

wherein the random numbers contained in the data set package serve to cryptographically bind the signed data set to a concretely observed state of the satellite navigation system, so that upon a subsequent authenticity verification by comparison of the random numbers contained in the data set with random sequences archived by independent reference receivers, replay-based GNSS spoofing attacks through repetition of previously transmitted random sequences are detectable.

Group L - Device- and Person-Related Attribution

64. The recording device according to any one of claims 1 to 13, characterized in that the recording device is configured to perform an authentication of a user and to include a user identifier derived therefrom as a metadatum in the data packet of the second sector before the second signing, wherein

a) the user identifier is not back-computable to the data underlying the authentication,

b) the recording device is configured to delete the data underlying the authentication after generation of the identifier within the second sector, so that these do not leave the recording device, and

c) the second signature attests to the presence of a completed user authentication at the time of recording.

65. The recording device according to claim 64, characterized in that the authentication is a biometric authentication, wherein the second signature attests to the presence of a natural person, and wherein the user identifier is generated from biometric raw data collected using at least one of the following sensors: an image sensor configured as an RGB-IR sensor, a depth sensor, a fingerprint sensor, an iris scanner.

66. The recording device according to claim 64 or 65, characterized in that the recording device is configured to optionally sign recordings with or without user authentication, wherein the second signature indicates in a machine-readable manner whether an authentication was present at the time of recording.

Group M - External Companion Device and Optical Provenance Signals

67. A system, comprising:

a) a cryptographic camera for generating cryptographically signed recording data, wherein the cryptographic camera has at least one signature engine and at least one private key assigned thereto, and

b) an external companion device,

characterized in that the companion device is configured to generate a machine-readable provenance signal which is introduced during a recording into the image and/or audio area captured by the cryptographic camera, wherein the provenance signal comprises provenance data in encoded form that comprise at least a timestamp, geocoordinates, a device identifier of the companion device, and a cryptographic authentication code over these data.

68. The system according to claim 67, characterized in that the companion device is a commercially available smartphone with application software that, without additional hardware, collects the provenance data, cryptographically secures them, and outputs them as an optical and/or acoustic provenance signal.

69. The system according to claim 67 or 68, characterized in that the companion device displays a machine-readable optical pattern, in particular a QR code or DataMatrix code, on a screen, wherein the optical pattern fully encodes the provenance data, so that they are readable from a single image, and wherein the optical pattern is either

a) directly visible in the field of view of the cryptographic camera, or

b) projected onto an object to be recorded and captured by the cryptographic camera as a projection.

70. The system according to claim 69, characterized in that the screen is divided into spatially separated areas, of which a first area carries the code symbol and at least one further area carries color- or brightness-modulated signals for transmitting additional data and/or for synchronization, wherein the multi-color capability of the screen is exploited through binary or multi-level modulation of the color channels red, green, and blue to increase the symbol capacity.

71. The system according to any one of claims 67 to 70, characterized in that the companion device additionally or alternatively transmits the provenance data to the cryptographic camera by time-modulated light of an LED, in particular the flashlight LED of a smartphone or an LED of a dongle, wherein the modulation is performed as on-off keying and the modulated light is received by the camera as a reflection from an object in the scene and/or by direct incidence through the objective onto the image sensor.

72. The system according to any one of claims 67 to 71, characterized in that the companion device additionally or alternatively outputs the provenance data acoustically, in particular in the near-ultrasound range, via a loudspeaker, and the cryptographic camera captures the acoustic signal via a microphone.

73. The system according to any one of claims 67 to 72, characterized in that the external companion device comprises a dongle connected to a smartphone, which comprises at least one of the following:

a) a radio receiver for receiving external environmental radio signals, in particular in the form of a software-defined radio receiver,

b) a focused LED light source with a narrower beam angle than a smartphone flash LED,

c) a projection module for generating structured light patterns that serve as a data-carrying provenance signal and/or as a focusing aid pattern,

d) a piezo loudspeaker,

e) a dedicated cryptographic key.

74. The recording device according to any one of claims 1 to 13, characterized in that the recording device comprises a verification unit which is configured to:

a) detect a provenance signal introduced by an external companion device according to any one of claims 67 to 73 into the captured image and/or audio area and to decode the provenance data contained therein,

b) verify the cryptographic authentication code contained in the provenance data,

c) perform at least one plausibility check in which received time and/or location information is compared with device-internal time and/or location information and/or a user identifier or a device identifier is checked against an authorization list, and

d) output a cryptographically signed recording only if the checks according to b) and c) are successful.

75. The recording device according to claim 74, characterized in that upon failed verification of the provenance signal, the recording is discarded and not persistently stored, or is stored unsigned with a machine-readable annotation regarding the failed verification.

76. The recording device according to claim 74 or 75, characterized in that the authorization list comprises person-related, device-related, and/or role-related entries and is held locally in the recording device or updated via a secured data connection.

77. A method for documenting a critical tracking event in a supply chain, comprising the steps of:

a) authenticating a documentation officer at a companion device,

b) generating a provenance signal by the companion device with at least a timestamp, geocoordinates, a device identifier of the companion device, an identifier of the documentation officer, and a cryptographic authentication code,

c) introducing the provenance signal into the image and/or audio area captured by a cryptographic camera,

d) generating a recording by the cryptographic camera,

e) verification of the provenance signal in the cryptographic camera according to any one of claims 74 to 76, and

f) outputting a cryptographically signed recording only upon successful verification,

wherein the verified provenance data are deposited as metadata in the signed recording.

78. The method according to claim 77, characterized in that at each critical tracking event of the supply chain at least one recording according to claim 77 is generated, and that the generated signed recordings together with the associated provenance data are stored for fulfillment of regulatory traceability requirements and provided to a supervisory authority upon request.

79. A computer program product comprising program code means that, when executed on a portable electronic device, in particular a smartphone, cause the device to operate as a companion device according to any one of claims 67 to 73, by:

a) collecting data for authentication of a user and transforming them into a user identifier that is not back-computable to the underlying data,

b) capturing time and location data,

c) generating from these data a cryptographically secured provenance message, and

d) outputting the provenance message as an optical pattern on a screen and/or as a time-modulated light signal and/or as an acoustic signal.

80. A method for generating a provenance manifest for a recording using a system according to any one of claims 67 to 73 or a recording device according to any one of claims 74 to 76 in conjunction with a cryptographic provenance protocol, in particular C2PA, characterized in that in the provenance manifest an additional assertion under an application-specific namespace is included, which contains in a structured manner the provenance data received from the companion device and verified in the camera, including a machine-readable verification status of the provenance signal.

81. The method according to claim 80, characterized in that the provenance manifest further comprises a hash of the optical pattern contained in the image, so that the linkage between the visible pattern and the extracted provenance data is cryptographically verifiable.

82. The system according to any one of claims 67 to 81, characterized in that a companion device is configured to establish, at the time of provenance collection, at least one data connection to an external service and to collect connection metadata arising thereby that are cryptographically signed by the external service or a third party cooperating therewith, as a component of the provenance data.

83. The system according to claim 82, characterized in that the connection metadata comprise at least one of the following:

(a) an OCSP Stapled Response,

(b) one or more Signed Certificate Timestamps (SCTs),

(c) a response from a Time Stamping Authority (TSA),

(d) a TLS certificate chain,

(e) DNS resolution data.

84. The system according to any one of claims 67 to 83, characterized in that the companion device collects cellular network and/or radio environment data and includes them as a component of the provenance data, wherein the environment data comprise at least one of the following:

(a) cell identifiers of serving and/or neighboring cellular network cells,

(b) a propagation delay parameter for distance estimation relative to a base station,

(c) MAC addresses of visible Wi-Fi access points.

85. The system according to any one of claims 67 to 84, characterized in that the recording device or a downstream verifier is configured to decode a machine-readable provenance signal of a companion device contained in the recorded signal and to verify the provenance data contained therein against sensor data of the recording device or against independently available reference data, wherein the result of the verification can be included as a metadatum in a signature of the recording device or in a provenance manifest.

86. The system according to claim 85, characterized in that the verification comprises at least one of the following:

(a) a comparison of geocoordinates,

(b) a comparison of time references,

(c) a comparison of radio environment data,

(d) a verification of a cryptographic signature of the companion device against a deposited public key of the companion device.

87. The system according to any one of claims 67 to 86, characterized in that the system is also operable without an existing internet connection of the companion device, wherein a machine-readable provenance signal comprises at least a device identifier of the companion device and a cryptographic signature of the companion device and optionally contains time or location data.

88. The system according to claim 87, characterized in that upon restoration of an internet connection, provenance data collected by the companion device are retroactively enriched with connection metadata according to any one of claims 82 or 83 and transmitted to a Time Stamping Authority, wherein a timestamp signed by the Time Stamping Authority is included in an associated provenance manifest.

89. The system according to any one of claims 67 to 88, characterized in that a machine-readable provenance signal of the companion device is structured into a mandatory part and a supplementary part, wherein the mandatory part comprises at least a device identifier of the companion device and a cryptographic signature of the companion device, and wherein the supplementary part is bound to the mandatory part via a cryptographic hash value.

90. The system according to claim 89, characterized in that a private key of the companion device is held in a secure hardware element of the companion device.

91. The system according to claim 90, characterized in that access to the private key is released upon successful user authentication according to a predetermined security level.

92. The system according to any one of claims 67 to 91, characterized in that provenance data of the companion device are suitable for simultaneously binding a recording to at least three of the following dimensions:

a) an authenticated person,

b) an identified companion device,

c) a network connection evidenced by connection metadata,

d) a geographic location at a specific time,

e) a radio environment.

93. The system according to any one of claims 67 to 92, characterized in that the recording device or a downstream verifier assigns to each recording a trust level that reflects the number and type of provenance dimensions supplied by the companion device and successfully verified, and that an image or video output or a corresponding release takes place only if the trust level reaches or exceeds a configurable minimum trust level.

94. The system according to claim 93, characterized in that the provenance dimensions comprise at least three of the following:

a) a verified device identity of the companion device,

b) a match of geocoordinates and/or time references,

c) an authentication of a user of the companion device,

d) a match of radio environment data,

e) a consistency of a radio environmental fingerprint.

95. The system according to claim 93 or 94, characterized in that in the case of video recordings a trust level is cumulatively built up over a plurality of successive individual frames, and that upon the trust level falling below the minimum trust level during an ongoing recording, affected segments are marked or their output is suppressed.

Group N - Device-External Challenge-Controlled Object Illumination

96. An illumination device for data-carrying object illumination, comprising

a) a communication interface for receiving a challenge provided by a device-external entity, wherein the challenge comprises at least one parameter for defining a temporally varying illumination pattern,

b) a control unit which is configured to derive a temporally varying illumination pattern from the parameter, and

c) at least one modulatable light source for illuminating an object to be recorded or a scene with the illumination pattern,

wherein the illumination pattern is configured such that it is detectable in concurrently generated image or video data of a camera as an optical response pattern and is subsequently verifiable for consistency with the challenge.

97. A method for generating subsequently verifiable recording data, comprising the steps of:

a) providing a challenge by a device-external entity to an illumination device, wherein the challenge comprises at least one parameter for defining a temporally varying illumination pattern,

b) illuminating an object or a scene by means of the illumination device with the illumination pattern derived from the challenge,

c) recording the illuminated scene by means of a camera within a time window specified by or derivable from the challenge, and

d) subsequently verifying whether the recorded image or video data exhibit an optical response pattern consistent with the challenge.

97a. The method according to claim 97, characterized in that the temporal classification of the optical response pattern that is decisive for the proof of authenticity is performed on the basis of external timestamps assigned by a device-external entity and/or externally captured reception times, and is not derived exclusively from camera-internal time information.

98. The method according to claim 97, wherein the camera cryptographically signs the recording data and generates a provenance manifest, in particular a manifest in accordance with the C2PA standard.

99. The illumination device according to claim 96 or the method according to any one of claims 97 or 98, wherein the challenge comprises a seed value from which the temporally varying illumination pattern is derived deterministically or pseudo-randomly, so that the sequence is not predictable in advance by an attacker but is subsequently reconstructable.

100. The illumination device according to any one of claims 96 or 99 or the method according to any one of claims 97 to 99, wherein the illumination pattern is generated in the near-infrared range and/or as spatially structured light.

100a. The illumination device according to any one of claims 96 to 100, characterized in that the at least one modulatable light source comprises a laser light source, in particular a laser diode, a VCSEL source, or a VCSEL array.

101. The method according to any one of claims 97 to 100, wherein the subsequent verification according to step d) comprises a correlation between the optical response pattern observed in the image or video material and the temporal and/or spatial progression of the illumination pattern expected from the challenge.

102. The method according to any one of claims 97 to 101, wherein additionally a physical plausibility check of the response pattern is performed, in particular with respect to consistency of shadow casting, interreflections, and/or material properties with the specified illumination geometry.

102a. The method according to any one of claims 97 to 102, characterized in that within the scope of the physical plausibility check, interference-related patterns in the optical response pattern, in particular speckle patterns, are evaluated.

103. The method according to any one of claims 97 to 102, wherein the device-external entity digitally signs the challenge and the signature is stored or transmitted together with the recording data or associated provenance data.

104. The method according to any one of claims 97 to 103, wherein a cryptographic hash value of the challenge is deposited with at least one time anchoring service, in particular a Time Stamping Authority, a transparency log, and/or a blockchain.

105. A system for generating tamper-resistant audiovisual recordings, comprising

a) a device-external entity which is configured to generate a time-critical challenge, wherein the challenge comprises at least one parameter for defining a temporally varying illumination pattern and a time window,

b) an illumination device according to any one of claims 96, 99, or 100, which receives the challenge and illuminates an object and/or a scene accordingly, and

c) a camera which captures the illumination effect as part of image or video data and preferably cryptographically signs them,

wherein the system is configured to verify the recorded image or video data for the presence of an optical response pattern consistent with the challenge.

105a. The system according to any one of claims 105 to 107, characterized in that the device-external entity is configured to perform the temporal classification of the optical response pattern that is decisive for the proof of authenticity on the basis of external

timestamps and/or external reception times, without being dependent on a camera-internal time base.

105b. The system according to any one of claims 105 to 105a, characterized in that the device-external entity is assigned to a trust domain separate from the camera user and is not controlled by the camera user.

105c. The system according to any one of claims 105 to 105b, characterized in that the camera has an image sensor with rolling-shutter-induced row-wise or column-wise temporal staggering of exposure, and that the authenticity verification comprises a check of the consistency of the optical response pattern with this temporal staggering.

106. The system according to claim 105, wherein the camera is a camera with provenance proof, in particular a camera with C2PA support.

107. The system according to claim 105 or 106, wherein the camera is a sectored recording device according to any one of claims 1 to 13, and at least one illumination means is assigned to a second sector, but the image sensor is assigned to a first sector.

108. The system according to any one of claims 105 to 107, wherein the device-external entity is a service of a video conferencing platform and provides the challenge to at least one illumination device used in the context of a video conference.